

Quantum error mitigation for bosonic modes using virtual distillation

3A Research Internship

Leonardo Finocchiaro

May 14, 2025



CHALMERS
UNIVERSITY OF TECHNOLOGY

Abstract

Quantum technologies aim to gain control over systems described by quantum mechanics to harness their computing potential. Despite the flourishing academic and industrial research, quantum systems are still too noisy to implement useful algorithms. Instead of waiting for the noise to become small enough, which would allow for full quantum error correction protocols, quantum error mitigation aims to reduce errors in a computation without correcting them completely, thus using fewer resources. Some protocols have already proven their usefulness, and recently, Virtual Distillation has been proposed as an interesting technique for qubit systems. This work studies how the protocol can also be applied to bosonic systems by developing its mathematical description and assessing its robustness using numerical simulations.

Given some observable $\hat{O}$, Virtual Distillation uses several copies of a noisy state to estimate the expectation value of $\hat{O}$ with respect to a distilled version of the state, thus yielding a more precise result. This computation is done without actually creating the distilled state, hence the name "virtual". For the case of qubits no direct circuit has been found so far for more than two copies. In this work, we find a direct way to implement the protocol for three bosonic modes in the case of the number operator $\hat{O} = \hat{n}$ and a general quadrature such as $\hat{O} = \hat{q}$. Numerical simulations show that upon incorporating interleaved noise, the protocol does not break down, and allow for testing other input states to broaden the analysis.

Acknowledgments

This five month internship has been made possible thanks to the help and support of numerous persons, whom I would like to thank here.

First of all, I would like to express my sincere gratitude towards my supervisor Assoc. Prof. Giulia Ferrini for her guidance throughout the whole internship, both on the technical level for my main projects and on the personal one for precious advice on life in Sweden. My deepest thanks go also to Dr. Adithi Udupa for our valuable discussions on my main subject of research as well as the precious resources needed to understand its technicalities. I will keep in mind the fruitful meetings we had together with Dr. Axel Eriksson from the experimentalist team.

I would like to thank Dr. Oliver Hahn, Timo Hillmann as well as Dr. Henning Kirchberg for our exciting and fascinating discussions on all kind of subjects, which have undoubtedly contributed a lot to my understanding of the theoretical concepts and the academic world in general. This is also true of the discussions with the members of the "Continuous-Variable Theory and Resource" group, and the whole "Applied Quantum Physics" laboratory, where I felt fully integrated and appreciated.

My acknowledgment goes to Ms. Linda Brånell for her precious administrative assistance throughout the duration of the internship.

I address special thanks to Prof. Eleni Diamanti, who put me in contact with my supervisor and took care of answering many of my questions and uncertainties in spite of her busy time schedule.

My internship was made possible by the financial support of Nano Area of Advance funding program, directed by Prof. Janine Splettstösser, to whom I am very grateful.

Contents

Abstract	1
Acknowledgments	2
Contents	3
1 Introduction	4
1.1 Quantum computing	4
1.2 Overview of the report	5
2 Theoretical grounds	6
2.1 Quantum states, density matrices	6
2.2 Loss channels and Kraus operators	7
3 The Virtual Distillation protocol	9
3.1 Mathematical description	9
3.1.1 Derivation	9
3.1.2 Diagonalizing method	10
3.2 Practical implementation for 3-mode virtual distillation	11
3.2.1 The diagonalizing $\hat{\mathcal{F}}_3$ gate	11
3.2.2 Measuring the number operator	12
3.2.3 Measuring quadrature operators	13
4 Numerical analysis	15
4.1 Incoherent errors: losses	15
4.2 Coherent errors	17
4.3 Other input states	18
4.4 Dephasing errors	21
5 Conclusion	24
Bibliography	25
Appendices	27
A Properties of a loss channel	27
B Proof of diagonalizing formula	29
C Parallel project: No-go theorem for Gaussian conversion protocol	32

1 Introduction

1.1 Quantum computing

During the last few decades, plenty of experimental breakthroughs and theoretical research have contributed to a growing interest in quantum technologies [1], [2]. Nowadays our contemporary laptops and smartphones have processors which can perform billions of basic operations every second. Nonetheless, some problems are known to be hard to solve even with such a computational power, for example optimization problems with large inputs, or simulation of complex molecular systems. The concern in quantum technologies relies on their ability to solve some of these intractable problems in much more reasonable timescales, thus widening our technical abilities and computing tools [3]. Moreover, obtaining such a computing advantage raises security interest for governments, especially as quantum algorithms have been found capable of efficiently factoring prime numbers and thus breaking most of modern cryptographic protocols [4].

In order to achieve such an advantage, several different platforms have been designed, each relying on various physical phenomena and with their own characteristics and drawbacks. Even though the physics behind laser-cooled neutral atoms [5] and superconducting circuits [6] is very different, a common framework is able to encapsulate and describe all quantum systems in an abstract fashion. Thanks to this, computer scientists can develop theoretical tools, protocols, and algorithms, which do not depend on a specific implementation but could theoretically run on each platform.

On the physics side, progress has been remarkable in the last decades, getting closer and closer to a fully functional quantum processor. Among the challenges that are still to be faced, *decoherence* is a limiting factor on almost all platforms. When a quantum system is exposed to its environment, which can be Earth's magnetic field, thermal fluctuations, collisions with other particles, etc., the system loses information over time and eventually reaches a completely noisy state preventing further computations. One could think that isolating the system completely from its environment would be a solution, but in order to perform actual computations, the system needs to be prepared, addressed and measured, and thus have a contact with its surrounding world. This trade-off is what sets a limit on the size of the systems we can control nowadays, and this limit is for now below what could be useful to run computations [7]. In fact, even if an error happens with a small probability of say 0.1%, even the smallest algorithms require several thousands of operations whose overall success rate would only be $(1 - 0.001)^{5000} \sim 0.6\%$.

In the long run, *quantum error correcting* codes and protocols promise to completely solve this issue, by detecting and correcting faulty operations occurring during a computation [8]. Inspired by classical error correcting protocols, these techniques are more and more thought as the solution to reach a functional quantum processor. However, as in the classical realm, errors can only be corrected if they are sufficiently sparse and below a certain threshold [9]. This error threshold is not reached nowadays in state-of-the-art quantum platforms, especially as the number of constituents increases.

While waiting for full quantum error correction to be possible, the field of *quantum error mitigation* (QEM) has grown, and provides tools to reduce the errors in a quantum algorithm without completely erasing them [10]. Combining these techniques could be a way to get below the error threshold, or perform a sub-class of interesting quantum algorithms in what is known as the Noisy Intermediate-Scale

Quantum era (NISQ). Quantum error mitigation is thus considered very promising in the short and medium term.

QEM techniques regroup for example zero-noise extrapolation, probabilistic error cancellation and symmetry-based error suppression [11]. In the context of continuous-variable quantum computation, other techniques for error mitigation have been considered, such as symmetry expansion [12]. Recent work has introduced Virtual Distillation (VD) [13], or Error Suppression by Derangement [14], as a potential candidate for error mitigation protocols. A team from Google Research used the protocol in a physical setup to compare it to other error mitigation protocols [15]. The general idea is to use several copies of a faulty, or noisy, state that we want to measure, and perform the measurement with respect to a *distilled* version of the state to obtain a result closer to the desired one. The term "virtual" is used to emphasize the fact that the distilled state is never actually created, but the measured quantity is mathematically equivalent to a measurement of this better version. Both references [13, 14] address the case of general measurements on qubits, and a previous work [16] mentioned Virtual Cooling as a technique to improve measurements on thermal states of bosonic modes.

In this report, we will focus on how the protocol performs for general measurements on bosonic modes, taking inspiration from both Virtual Distillation and Virtual Cooling strategies. It turns out that bosonic modes allow for scaling up the number of copies used in the protocol, notably for $M = 3$ copies for which qubit Virtual Distillation is only known to be possible with an approximate circuit. We managed to show the performance of the protocol in the case of number operator and quadrature measurements, which are of great interest for the study of bosonic systems.

We chose to focus on a specific approach called *diagonalizing technique*, as opposed to the *auxiliary qubit* approach, which turned out to be less suited for applications to bosonic modes. This other approach relied on hybrid controlled-SWAP gates, where a qubit controls two bosonic modes. State of the art platforms developed at Chalmers did not seem to be suited for this kind of operations, and this is why we turned our attention on the first technique.

1.2 Overview of the report

Throughout the report, we will explore the theoretical foundations of quantum mechanics that allow to clearly and compactly formulate the protocol under study. The goal is not to explain in full detail the whole theory of quantum physics, we choose instead an approach inspired by quantum information theory. In this framework, quantum states and operations are described through abstract objects belonging to and acting on a Hilbert space. In particular, to model evolution over time, the main formulas will be stated without full proof. We deliberately choose to put aside Schrödinger's equation and Lindblad theory to focus on the description of the Virtual Distillation protocol and its performance through numerical simulations.

After having introduced all the relevant objects used in the following, we will describe the Virtual Distillation protocol for bosonic modes, both in its mathematical formulation and with an experimental way to perform it using standard operations such as beam splitters and number operator measurements. We will then look at a numerical implementation, to confirm the theoretical results, assess the robustness of the protocol when several sources of errors are added, and try varying the input parameters.

All the code used to generate the plots and to perform the calculations can be found on GitHub <https://github.com/lfinocchiaro/virtual-distillation>.

2 Theoretical grounds

2.1 Quantum states, density matrices

The state of a quantum object is described through the formalism of Hilbert spaces. A Hilbert space $\mathcal{H}$ is a complex vector space, endowed with a Hermitian product, and complete, *i.e.* satisfying some properties of convergence. This formalism immediately encapsulates the two most important features of a quantum system: *superposition*, which arises from the vector space structure, and *entanglement*, which is a consequence of taking the tensor product of Hilbert spaces when describing several systems. The elements of $\mathcal{H}$, the possible states of our system, are usually described using Dirac notation $|\psi\rangle$, which allows for simply writing down a state and its dual $\langle\psi|$, the associated linear form.

Quantum computing and information theory distinguish two paradigms. *Discrete variable quantum computing* focuses on systems called *qudits* described by a finite dimensional Hilbert space. The simplest case and also the most prominent in actual research is the *qubit* case, corresponding to a two-dimensional space. Several platforms implement qubits, most notably cold atoms, trapped ions, NV-centers and superconducting circuits. The two-dimensional nature of their Hilbert space comes from a reduced set of energy levels addressed in the experiments (for example a ground and an excited state for the valence electron of an atom). We often choose a convenient basis of the space noted $(|0\rangle, |1\rangle)$, and define operators acting on the space such as the Pauli operators $\hat{X}, \hat{Y}, \hat{Z}$. In this case, the operators are linear maps of a two-dimensional Hilbert space and can thus be fully written down as 2×2 complex matrices, and the states as 2-dimensional normalized complex column vectors. Qudits also receive some interest and are experimentally interesting because they generalize the concept of qubits, even though they are harder to control with high precision the more dimensions there are. The operators acting on a d -dimensional Hilbert space are described by $d \times d$ matrices after choosing an appropriate basis $|0\rangle, \dots, |d-1\rangle$.

The second paradigm is called *continuous variable* or *bosonic* quantum computing, and corresponds to the case where the Hilbert space is infinite dimensional with the structure of a quantized harmonic oscillator. This is a common feature found in nature, when the energy levels of a system are infinite and evenly spaced. In this framework, the whole Hilbert space is generated by the vacuum state $|0\rangle$ and a *creation operator* $\hat{a}^\dagger$, which allows for recursively creating an infinite set of orthogonal states called *Fock states* forming a basis of $\mathcal{H}$:

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle.$$

The structure of the space comes from the *commutation relation*

$$[\hat{a}, \hat{a}^\dagger] = \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = \mathbb{1}.$$

These systems are referred as harmonic oscillators because the usual concepts of position and momentum of an oscillator are encapsulated in two Hermitian operators (or *observables* in the language of quantum mechanics) called *quadratures*:

$$\hat{q} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}} \quad \text{and} \quad \hat{p} = \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}}.$$

Probably the simplest quantum object described by such a Hilbert space is light, or more precisely one mode of the quantized electro-magnetic field, whose fundamental excitations are called *photons*. The

experimental setup usually consists of a cavity with two mirrors, trapping the electro-magnetic field at a desired frequency and allowing, for small enough temperatures, for reaching a quantum behavior. Other quantum systems are also described by a quantized harmonic oscillator Hilbert space, such as micro-wave cavities or mechanical oscillators. We call these systems continuous-variable systems because of the continuous spectrum of the observables $\hat{q}$, $\hat{p}$. In the following we will focus on this description mostly forgetting the actual underlying implementation.

The final object we will use throughout this report is the density matrix. When unpredictable phenomena occur, such as errors or external measurements (whose outcomes are genuinely random according to Quantum Mechanics), information about the state is lost and we can't say anymore in which state of the Hilbert space, or *pure state*, the quantum system is exactly. To represent a probabilistic mixture we use a *density matrix*, representing a *mixed state*, and defined as

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|,$$

where $|\psi_i\rangle$ are orthogonal pure states and $p_i \in [0, 1]$ the (classical) probability that the system is in such a state. It is useful to keep in mind that this only represents a lack of information, and that the actual physical system is always in a pure state $|\psi_i\rangle$, but we don't know which one.

In the case of qudits and finite dimensional Hilbert spaces, $\hat{\rho}$ can be simply written as a $d \times d$ matrix, but in the context of continuous-variable quantum information, this description is not possible anymore and $\hat{\rho}$ has the structure of a Hermitian operator.

When a measurement is performed on a quantum system, it is generally described by an *observable* $\hat{O}$, a Hermitian operator of the Hilbert space. Its expected value (*i.e.* its average value over a lot of repetitions) is given by $\langle\psi|\hat{O}|\psi\rangle$ in the case of a pure state $|\psi\rangle$, and by $\text{Tr}(\hat{\rho}\hat{O}) = \sum p_i \langle\psi_i|\hat{O}|\psi_i\rangle$ in the case of a mixed state $\hat{\rho}$.

The states we will extensively use in the following are

- Fock states $|0\rangle, |1\rangle, \dots, |n\rangle, \dots$
- Coherent states $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ for $\alpha \in \mathbb{C}$.
- The binomial state $\frac{|0\rangle + |4\rangle}{\sqrt{2}}$
- The GKP state $|0\rangle_{\text{GKP}}$, which has received a lot of interest due to its practical use for encoding a qubit in a bosonic mode. A detailed description is given section 4.3.

2.2 Loss channels and Kraus operators

Most physical systems are subject to noise, which means that by interfering with the environment, they may change uncontrollably over time, losing their original information. This is called *decoherence* and is the main limiting factor in our modern physical realization of quantum computers.

In order to model a general noise channel responsible for decoherence, Kraus operators are a useful tool. The general evolution of a state is described by the generalized Schrödinger equation called the Lindblad equation. Without entering in the details, we will here simply state the main results of the theory that will be useful for us (see [17] Appendix F for a detailed derivation).

In the case of a pure loss channel, the associated Kraus operators are

$$\hat{A}_k(t) = \frac{(1 - e^{-\kappa t})^{k/2}}{\sqrt{k!}} e^{-\kappa \hat{n}t/2} \hat{a}^k \quad \text{for } k \in \mathbb{N}. \quad (1)$$

The resulting mixed state after applying a loss channel to a state $\hat{\rho}_0$ for a given time t writes

$$\hat{\rho}(t) = \sum_{k=0}^{\infty} \hat{A}_k(t) \hat{\rho}_0 \hat{A}_k^\dagger(t). \quad (2)$$

The main results used in the following are stated below, and proven in detail in Appendix A.

1. The noisy state is still a valid density matrix, *i.e.* $\text{Tr}(\hat{\rho}(t)) = 1$.
2. We have that $\langle \hat{n} \rangle(t) = e^{-\kappa t} \langle \hat{n} \rangle(0)$, showing that the average photon number follows an exponential decay, with a decay rate κ , as expected by the intuitive interpretation of a loss channel.
3. The explicit form for a time-evolved coherent state given $\hat{\rho}_0 = |\alpha\rangle\langle\alpha|$ writes

$$\hat{\rho}(t) = \left| \alpha e^{-\kappa t/2} \right\rangle \left\langle \alpha e^{-\kappa t/2} \right|. \quad (3)$$

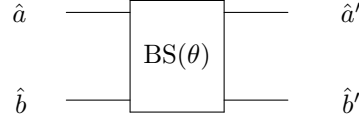
4. For an initial Fock state $\hat{\rho}_0 = |n\rangle\langle n|$, we have

$$\hat{\rho}(t) = e^{-\kappa n t} \sum_{k=0}^n \binom{n}{k} (e^{\kappa t} - 1)^k |n-k\rangle\langle n-k|. \quad (4)$$

In the last case, the resulting mixed state has (for small κt) a dominant eigenvector $|n\rangle$, and as other eigenvectors some orthogonal Fock states of smaller photon number $|n-k\rangle$. A more exhaustive analysis is performed in section 4.1.

Commutation between losses and linear operations

An interesting result that we found useful in our analysis is that in the case of a linear operation, like the beam-splitters we will consider in the following, all the losses could be thought as happening at the beginning. Consider for instance a unitary beam-splitter operation acting on the annihilation operators as $\begin{pmatrix} \hat{a}' \\ \hat{b}' \end{pmatrix} = \text{BS}(\theta) \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix}$, where $\text{BS}(\theta) = \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}$.



A loss channel as introduced above and acting after the gate can be seen as transforming the annihilation operators as $\hat{a}' \rightarrow \eta \hat{a}'$ and $\hat{b}' \rightarrow \eta \hat{b}'$, with $\eta = e^{\kappa t/2}$. By linearity, this amounts to transform $\begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \text{BS}(\theta)^{-1} \begin{pmatrix} \hat{a}' \\ \hat{b}' \end{pmatrix} \rightarrow \eta \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix}$, meaning that we can as well consider that this loss happened at the beginning. Here we can explicitly compute the loss factor η using our model for losses (some results from Appendix A are needed) :

$$\sum_{k=0}^{\infty} \hat{A}_k \hat{a} \hat{A}_k^{\dagger} = \sum_{k=0}^{\infty} \frac{(1 - e^{-\kappa t})^k}{k!} e^{-\kappa t \hat{n}/2} \hat{a}^{k+1} (\hat{a}^{\dagger})^k e^{-\kappa t \hat{n}/2} = \hat{a} e^{\kappa t/2} \underbrace{\sum_{k=0}^{\infty} \hat{A}_k \hat{A}_k^{\dagger}}_{\substack{\mathbb{1} \\ \text{(see below)}}} = \underbrace{e^{\kappa t/2}}_{\eta} \hat{a},$$

$$\begin{aligned} \text{using that} \quad \sum_{k=0}^{\infty} \hat{A}_k \hat{A}_k^{\dagger} &= \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} e^{-\gamma \hat{n}/2} \underbrace{\hat{a}^k \hat{a}^{\dagger k}}_{\hat{n}(\hat{n}+1)\dots(\hat{n}+k-1)} e^{-\gamma \hat{n}/2} \\ &= e^{-\gamma \hat{n}} \sum_{k=0}^{\infty} \binom{\hat{n} + k - 1}{k} (1 - e^{-\gamma})^k = e^{-\gamma \hat{n}} \frac{1}{(1 - (1 - e^{-\gamma}))^{\hat{n}}} = \mathbb{1}, \end{aligned}$$

where we used the negative binomial formula $\sum_{k=0}^{\infty} \binom{N-1+k}{k} x^k = (1-x)^{-N}$.

To conclude, if a quantum circuit is only made of linear operations such as beam splitters, all the losses in the middle of the circuit can be thought as happening at the beginning. This simplification will allow for easily taking this kind of *interleaved errors* into account during the protocol.

3 The Virtual Distillation protocol

3.1 Mathematical description

This work is inspired by [16] and [13], respectively introducing a "Virtual Cooling" technique for measuring observables on thermal states of bosonic system, and a "Virtual Distillation" protocol on qubits, which we here transpose to bosonic systems. It turns out that most of the expressions stay the same as in the qubit case, but the diagonalizing circuit is made possible using techniques from Virtual Cooling.

3.1.1 Derivation

We have an observable $\hat{O}$ of which we want to measure the average value on a pure state $|\psi\rangle$

$$\langle\psi|\hat{O}|\psi\rangle. \quad (5)$$

We only however have access to noisy states $\hat{\rho}$, close to $|\psi\rangle$, *i.e.* of the form

$$\hat{\rho} = (1-p)|\psi\rangle\langle\psi| + \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad (6)$$

with $p = \sum_i p_i$. This decomposition is written in terms of eigenvectors of $\hat{\rho}$, and we suppose that $|\psi\rangle$ is the dominant one (the one with highest probability). In particular, the states $|\psi\rangle, |\psi_i\rangle$ are orthogonal.

In this situation, we can consider from a theoretical point of view the following "purified" state:

$$\hat{\hat{\rho}} \stackrel{\text{def}}{=} \frac{\hat{\rho}^M}{\text{Tr}(\hat{\rho}^M)} = \frac{(1-p)^M |\psi\rangle\langle\psi| + \sum_i p_i^M |\psi_i\rangle\langle\psi_i|}{(1-p)^M + \sum_i p_i^M}.$$

We call this state "purified" because its dominant eigenvector is still $|\psi\rangle$, but now with a much higher precision:

$$\langle\psi|\hat{\hat{\rho}}|\psi\rangle = \frac{(1-p)^M}{(1-p)^M + \sum_i p_i^M} \sim 1 - p^M > 1 - p = \langle\psi|\hat{\rho}|\psi\rangle.$$

In order to measure the desired quantity Eq. (5), this new purified state would yield a better result than the original noisy state $\hat{\rho}$, which approaches the ideal value for large values of M :

$$\langle\psi|\hat{O}|\psi\rangle = \lim_{M \rightarrow \infty} \text{Tr}(\hat{O}\hat{\hat{\rho}}) = \lim_{M \rightarrow \infty} \frac{\text{Tr}(\hat{O}\hat{\rho}^M)}{\text{Tr}(\hat{\rho}^M)}.$$

In the Virtual Distillation protocol, we don't explicitly build the state $\hat{\hat{\rho}}$, but rather use several copies of $\hat{\rho}$ to estimate the two quantities $\text{Tr}(\hat{O}\hat{\rho}^M)$ and $\text{Tr}(\hat{\rho}^M)$. To do this, we use a formula connecting the abstract object $\hat{\rho}^M$ to a physically meaningful state $\hat{\rho}^{\otimes M}$:

$$\text{Tr}(\hat{O}\hat{\rho}^M)_{(\mathcal{H})} = \text{Tr}(\hat{O}^{(M)}\hat{S}^{(M)}\hat{\rho}^{\otimes M})_{(\mathcal{H}^M)}, \quad (7)$$

where we introduced the two operators $\hat{O}^{(M)}$ "symmetrized $\hat{O}$ " and $\hat{S}^{(M)}$ "shift" on $\mathcal{H}^M$:

$$\hat{O}^{(M)} = \frac{1}{M} \left(\hat{O} \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1} + \mathbb{1} \otimes \hat{O} \otimes \dots \otimes \mathbb{1} + \dots + \mathbb{1} \otimes \dots \otimes \mathbb{1} \otimes \hat{O} \right),$$

$$\hat{S}^{(M)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_M\rangle = |\psi_2\rangle \otimes \dots \otimes |\psi_M\rangle \otimes |\psi_1\rangle.$$

This way, we are left with computing the quantities $\text{Tr} \left(\hat{O}^{(M)} \hat{S}^{(M)} \hat{\rho}^{\otimes M} \right)$ and $\text{Tr} \left(\hat{S}^{(M)} \hat{\rho}^{\otimes M} \right)$, *i.e.* the average values of $\hat{O}^{(M)} \hat{S}^{(M)}$ and $\hat{S}^{(M)}$ on the state $\hat{\rho}^{\otimes M}$. Eq. (7) can be proven by writing down explicitly the traces as sum over components (there exist a more visual proof using tensor networks shown in [13]), where $\hat{O}^1$ is the first term of $\hat{O}^{(M)}$:

$$\begin{aligned} \text{Tr}(\hat{O}^1 \hat{S}^{(M)} \hat{\rho}^{\otimes M}) &= \sum_{i_1, \dots, i_M, j_1, \dots, j_M, k_1, \dots, k_M} (O_{k_1, j_1} \delta_{k_2, j_2} \dots \delta_{k_M, j_M}) (\delta_{i_2, j_1} \dots \delta_{i_1, j_M}) (\rho_{j_1, k_1} \dots \rho_{j_M, k_M}) \\ &= \sum_{i_1, \dots, i_M, j_1, \dots, j_M, k} O_{k, i_1} \delta_{i_2, j_1} \dots \delta_{i_1, j_M} \rho_{j_1, k} \rho_{j_2, i_2} \dots \rho_{j_M, i_M} \\ &= \sum_{i_1, \dots, i_M, k} O_{k, i_1} \rho_{i_1, i_M} \rho_{i_M, i_{M-1}} \dots \rho_{i_2, k} \\ &= \text{Tr}(\hat{O} \hat{\rho}^M). \end{aligned} \tag{8}$$

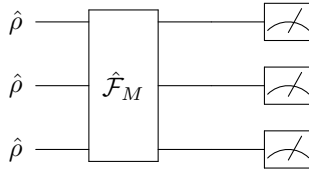
Eq. (7) is then obtained by symmetry and linearity.

The protocol is expected to work when the input state is *close* to the target pure state, in the sense that its dominant eigenvector is the pure state itself. In the case of a *coherent drift*, meaning that the noisy input state has a dominant eigenvector $|\psi'\rangle$ different from $|\psi\rangle$, the protocol is expected to distil with respect to $|\psi'\rangle$. Most notably if the input state is a coherent state $|\alpha\rangle$ which endured some loss, our previous analysis in Eq. (3) shows that the noisy state is not of the desired form because its dominant eigenvector has drifted, so the protocol is not expected to yield an advantage. Conversely, the noisy Fock state is exactly in the appropriate form because its dominant eigenvector is still $|n\rangle$ and all other contributions created by loss are orthogonal to it.

In [18], it is proven that the coherent drift in the eigenvector of a state is negligible compared to the incoherent loss in amplitude, in "relevant scenarios" defined as those where $|\psi\rangle$ is an eigenstate of $\hat{O}$. For this reason, we will focus on applying this algorithm to $\hat{O} = \hat{n}$ the number operator, on noisy copies of $|\psi\rangle = |n\rangle$ a Fock state, but we will also establish a result for $\hat{O} = \hat{q}$ (or any generalized quadrature).

3.1.2 Diagonalizing method

This method to compute the desired quantities consists in finding a gate that diagonalizes the two operators $\hat{O}^{(M)} \hat{S}^{(M)}$ and $\hat{S}^{(M)}$ in a basis easy to measure, as described in the circuit below for the case $M = 3$:



We here state a relationship claimed in [16] but for which no proof was given. In appendix B we provide a complete proof which we found using basic linear algebra and calculus tools.

Let's consider M bosonic modes with associated ladder operators $\hat{a}_p$ and number operators $\hat{n}_p = \hat{a}_p^\dagger \hat{a}_p$. $\hat{S}^{(M)}$ is the shift operator as defined in the previous subsection.

Proposition 1. Define the unitary $\hat{\mathcal{F}}_M$ called "discrete Fourier transform" such that

$$\hat{\mathcal{F}}_M^\dagger \hat{a}_p \hat{\mathcal{F}}_M = \frac{1}{\sqrt{M}} \sum_{k=1}^M e^{2i\pi \frac{kp}{M}} \hat{a}_k \quad \text{for } p \in \{1, \dots, M\}.$$

If we consider the (unitary) operator $\hat{\mathcal{R}}_M$, defined as

$$\hat{\mathcal{R}}_M = \exp \left\{ 2i\pi \frac{1}{M} \sum_{p=1}^M p \hat{n}_p \right\} = \bigotimes_{p=1}^M e^{\frac{2i\pi}{M} p \hat{n}_p},$$

then we have the following identity:

$$\hat{\mathcal{F}}_M^\dagger \hat{\mathcal{R}}_M \hat{\mathcal{F}}_M = \hat{S}^{(M)}. \quad (9)$$

Note that $\hat{\mathcal{R}}_M$ is a tensor product of operators acting on only one mode at a time. If we are able to measure the number operator on each mode at the end of the protocol, we can access the expectation value of $\hat{\mathcal{R}}_M$, and thus directly retrieve the desired quantity $\text{Tr} \left(\hat{S}^{(M)} \right)$, which is half of what we need. Now the difficult part is to try and implement such a diagonalizing gate $\hat{\mathcal{F}}_M$, and after that to measure the other quantity $\text{Tr} \left(\hat{O}^{(M)} \hat{S}^{(M)} \right)$ using this setup. As we will see, in the case $\hat{O} = \hat{n}$ and $\hat{O} = \hat{q}$, this second quantity is immediately retrieved by a similar formula as the first. For more general $\hat{O}$, a more complicated measurement may be needed, and the $\hat{\mathcal{F}}_3$ gate may not be as useful as here.

3.2 Practical implementation for 3-mode virtual distillation

3.2.1 The diagonalizing $\hat{\mathcal{F}}_3$ gate

The $\hat{\mathcal{F}}_M$ gate introduced above is also known as a "multiport beam-splitter" and some references give practical ways to implement it [19, 20].

Using the definition of $\hat{\mathcal{F}}_M$, we know that

$$\hat{\mathcal{F}}_M^\dagger \begin{pmatrix} \hat{a}_1 \\ \vdots \\ \hat{a}_M \end{pmatrix} \hat{\mathcal{F}}_M = \frac{1}{\sqrt{M}} \begin{pmatrix} \omega & \omega^2 & \dots & \omega^M \\ \omega^2 & \omega^4 & \dots & \omega^{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \omega^M & \dots & \omega^{M^2} \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \vdots \\ \hat{a}_M \end{pmatrix}.$$

Let's tackle the case of $M = 3$ copies of a single mode state. One way to obtain the $\hat{\mathcal{F}}_3$ gate is to perform three successive beam splitter operations between the modes, as well as some phase shifts. The general action of a beam splitter $\text{BS}_{12}(\theta)$ on the annihilation operators of two bosonic modes 1 and 2 can be written as

$$\text{BS}_{12}(\theta) = \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}.$$

The generator of these transformation is the Hamiltonian $\hat{H}_{BS} = \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1$, which means that we have

$$e^{-i\theta \hat{H}_{BS}} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} e^{i\theta \hat{H}_{BS}} = \text{BS}_{12}(\theta) \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix}.$$

Note that some references introduce $\hat{H}'_{BS} = i(\hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1)$ which yields a different form for the beam splitter transformation (the usual rotation matrix).

A phase shift is described by a single-mode Hamiltonian $\hat{H}_{PS} = \hat{n}$, acting as

$$e^{-i\phi \hat{H}_{PS}} \hat{a} e^{i\phi \hat{H}_{PS}} = e^{i\phi} \hat{a}.$$

In micro-wave experiments, it is often possible to incorporate phase shifts into the beam splitting operation.

We can now consider our 3-mode system, and perform three beam splitter operations among them:

$$\begin{aligned}
 \text{BS}_{23} \left(\sin \theta_1 = \frac{1}{\sqrt{2}} \right) \text{BS}_{13} \left(\sin \theta_2 = \frac{1}{\sqrt{3}} \right) \text{BS}_{12} \left(\sin \theta_3 = \frac{1}{\sqrt{2}} \right) &= \\
 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{\frac{2}{3}} & 0 & \frac{i}{\sqrt{3}} \\ 0 & 1 & 0 \\ \frac{i}{\sqrt{3}} & 0 & \sqrt{\frac{2}{3}} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & i & i \\ e^{\frac{2i\pi}{3}} & e^{-\frac{i\pi}{6}} & i \\ e^{\frac{5i\pi}{6}} & e^{\frac{2i\pi}{3}} & 1 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \omega & \omega^2 & 1 \\ \omega^2 & \omega & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} e^{\frac{5i\pi}{6}} & 0 & 0 \\ 0 & e^{\frac{2i\pi}{3}} & 0 \\ 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

The two outermost matrices correspond to phase-shifts acting on one mode at a time, and the middle one is the action we expect for our gate. By taking these phase shifts into account, we thus retrieve an explicit way to implement the $\hat{\mathcal{F}}_3$ gate, with 3 beam-splitters and 4 phase-shifters of the form $e^{i\phi\hat{n}}$:

$$\hat{\mathcal{F}}_3 = e^{-\frac{i\pi}{2}\hat{n}_1 - \frac{i\pi}{2}\hat{n}_2} \cdot e^{\frac{i\pi}{4}(\hat{a}_2^\dagger\hat{a}_3 + \hat{a}_3^\dagger\hat{a}_2)} \cdot e^{i\arcsin(\frac{1}{\sqrt{3}})(\hat{a}_1^\dagger\hat{a}_3 + \hat{a}_3^\dagger\hat{a}_1)} \cdot e^{\frac{i\pi}{4}(\hat{a}_1^\dagger\hat{a}_2 + \hat{a}_2^\dagger\hat{a}_1)} \cdot e^{-\frac{5i\pi}{6}\hat{n}_1 - \frac{2i\pi}{3}\hat{n}_2}. \quad (10)$$

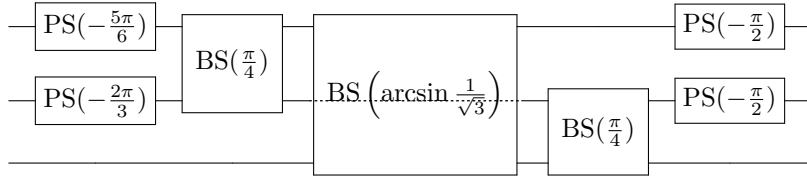


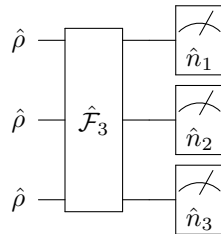
Figure 1: Full circuit implementing the diagonalizing $\hat{\mathcal{F}}_3$ gate. Phase shifters noted as $\text{PS}(\phi)$ act on one mode, and beam splitters noted $\text{BS}(\theta)$ act on two modes. The middle gate only acts on the top and bottom modes.

The corresponding gate implementation is shown in Figure 1. In references [19, 20], a similar construction is established in the case of the 4-mode multiport beam splitter $\hat{\mathcal{F}}_4$ using four beam splitters and (at least) one phase shifter. They also provide a theoretical framework for $M \geq 5$ but with no physical circuit. In the following, we decided to focus on 3-mode Virtual Distillation in order to provide numerical simulations (which become quickly intractable the more modes there are).

3.2.2 Measuring the number operator

Because of the convenient shape of the $\hat{\mathcal{R}}_M$ operator in Eq.(9), a natural observable to consider is the *number operator* $\hat{n}$. We thus establish the protocol in the case of $\hat{O} = \hat{n}$, and in the case of $M = 3$ modes as started in the previous section.

This situation is simple because as we will see, applying the diagonalizing $\hat{\mathcal{F}}_3$ gate and then measuring in the $\hat{n}$ basis is enough to retrieve the two values $\langle \hat{S}^{(3)} \rangle$ and $\langle \hat{n}^{(3)} \hat{S}^{(3)} \rangle$ at the same time. We consider the circuit depicted below:



The state of the system before measurement is $\tilde{\rho} = \hat{\mathcal{F}}_3 \rho^{\otimes 3} \hat{\mathcal{F}}_3^\dagger$. We can express the quantities $\langle \hat{S}^{(3)} \rangle$

and $\langle \hat{n}^{(3)} \hat{S}^{(3)} \rangle$ using Eq. (9) in terms of this state:

$$\langle \hat{S}^{(3)} \rangle = \text{Tr} \left(\hat{S}_3 \rho^{\otimes 3} \right) = \text{Tr} \left(\hat{\mathcal{F}}_3^\dagger \hat{\mathcal{R}}_3 \hat{\mathcal{F}}_3 \rho^{\otimes 3} \right) = \text{Tr} \left(\hat{\mathcal{R}}_3 \hat{\rho} \right) = \left\langle \exp \left(\frac{2i\pi}{3} (\hat{n}_1 + 2\hat{n}_2) \right) \right\rangle, \quad (11)$$

$$\langle \hat{n}^{(3)} \hat{S}^{(3)} \rangle = \text{Tr} \left(\hat{n}^{(3)} \hat{S}_3 \rho^{\otimes 3} \right) = \text{Tr} \left(\left(\hat{\mathcal{F}}_3 \hat{n}^{(3)} \hat{\mathcal{F}}_3^\dagger \right) \hat{\mathcal{R}}_3 \hat{\rho} \right) = \left\langle \frac{\hat{n}_1 + \hat{n}_2 + \hat{n}_3}{3} e^{\frac{2i\pi}{3} (\hat{n}_1 + 2\hat{n}_2)} \right\rangle, \quad (12)$$

where we used the general fact that

$$\begin{aligned} \hat{\mathcal{F}}_M \hat{n}^{(M)} \hat{\mathcal{F}}_M^\dagger &= \frac{1}{M} \sum_{p=1}^M \left(\hat{\mathcal{F}}_M \hat{a}_p^\dagger \hat{\mathcal{F}}_M^\dagger \right) \left(\hat{\mathcal{F}}_M \hat{a}_p \hat{\mathcal{F}}_M^\dagger \right) \\ &= \frac{1}{M^2} \sum_{p,k,l=1}^M e^{\frac{2i\pi}{M} kp} e^{-\frac{2i\pi}{M} lp} \hat{a}_k^\dagger \hat{a}_l \quad \text{using the inverse formula } \hat{\mathcal{F}}_M \hat{a}_p \hat{\mathcal{F}}_M^\dagger = \frac{1}{\sqrt{M}} \sum_{k=1}^M e^{-2i\pi \frac{kp}{M}} \hat{a}_k \\ &= \frac{1}{M} \sum_{k=1}^M \hat{a}_k^\dagger \hat{a}_k \quad \text{because } \sum_p (\omega^{k-l})^p = M \delta_{kl} \\ &= \hat{n}^{(M)} \end{aligned}$$

At this point, we can repeat the experiment and have an estimate of the two mean values (11) and (12). Taking the ratio will then give an estimate of the mean value of $\hat{n}$ on the purified state.

Expected results for Fock state input with losses

We here take a toy-model to illustrate the protocol, and we will later develop a more advanced one to better fit the experimental setup. Suppose that the initial state ρ is close to a target Fock state $|n\rangle$, but may have endured a single loss with probability $p \ll 1$. This situation is described by the density matrix:

$$\rho = (1-p) |n\rangle\langle n| + p |n-1\rangle\langle n-1|. \quad (13)$$

If we simply measure the number operator on a single mode, we would get an expected value $\text{Tr}(\hat{n}\rho) = n-p$. Here instead we start with three copies, apply the $\hat{\mathcal{F}}_3$ gate (with three beam splitters for example), and finally measure each mode in the number state. We then compute the two expressions (11) and (12), averaging them for every repetition. Taking their ratio at the end should give the quantity

$$\frac{\text{Tr}(\hat{n}\rho^3)}{\text{Tr}(\rho^3)} = \frac{n(1-p)^3 + (n-1)p^3}{(1-p)^3 + p^3} = n - \frac{p^3}{(1-p)^3 + p^3} \sim n - p^3$$

It is worth noting that if we simply computed the average number operator on three noisy copies *without* performing the $\hat{\mathcal{F}}_3$ gate, we would get no benefit at all:

$$\text{Tr}(\hat{n}^{(3)} \rho^{\otimes 3}) = (1-p)^3 n + 3(1-p)^2 p \left(n - \frac{1}{3} \right) + 3(1-p)p^2 \left(n - \frac{2}{3} \right) + p^3(n-1) = n - p.$$

When working with Fock states, which are symmetric in phase-space, dephasing errors have no effect on the states as they only affect the angular coordinate. We will come back to dephasing errors in section 4.4 as we will study other input states which are not symmetric.

3.2.3 Measuring quadrature operators

As it turns out, the formalism we have developed allows for generalizing these results to measurements of other observables such as the quadrature operators $\hat{q}$ and $\hat{p}$ defined as

$$\hat{q} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{p} = \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}}.$$

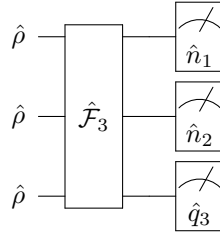
We use the same setup with the $\hat{\mathcal{F}}_3$ gate and the only difference is the final measurement. As in Eq. (12), we need to compute the following quantity:

$$\begin{aligned}
 \hat{\mathcal{F}}_M \hat{q}^{(M)} \hat{\mathcal{F}}_M^\dagger &= \frac{1}{\sqrt{2M}} \sum_{j=1}^M \hat{\mathcal{F}}_M (\hat{a}_j + \hat{a}_j^\dagger) \hat{\mathcal{F}}_M^\dagger \\
 &= \frac{1}{\sqrt{2M}} \sum_{j=1}^M \left(\frac{1}{\sqrt{M}} \sum_{k=1}^M \omega^{-kj} \hat{a}_k + \frac{i}{\sqrt{M}} \sum_{k=1}^M \omega^{kj} \hat{a}_k^\dagger \right) \\
 &= \frac{1}{\sqrt{2}\sqrt{M}} \sum_{k=1}^M \left[\left(\frac{1}{M} \sum_{j=1}^M \omega^{-kj} \right) \hat{a}_k + i \left(\frac{1}{M} \sum_{j=1}^M \omega^{kj} \right) \hat{a}_k^\dagger \right] \\
 &= \frac{1}{\sqrt{2M}} (\hat{a}_M + i \hat{a}_M^\dagger) = \frac{1}{\sqrt{M}} \hat{q}_M.
 \end{aligned}$$

For the case $M = 3$, we thus obtain that

$$\text{Tr}(\hat{q} \hat{\rho}^3) = \text{Tr}(\hat{q}^{(3)} \hat{S}^{(3)} \rho^{\otimes 3}) = \text{Tr}\left(\frac{\hat{q}_3}{\sqrt{3}} \hat{\mathcal{R}}_3 \tilde{\rho}\right),$$

and the operator $\frac{\hat{q}_3}{\sqrt{3}} \hat{\mathcal{R}}_3 = \frac{\hat{q}_3}{\sqrt{3}} e^{\frac{2i\pi}{3}(\hat{n}_1 + 2\hat{n}_2)}$ can be measured by directly measuring $\hat{n}_1$, $\hat{n}_2$ and $\hat{q}_3$:



If we perform the same calculation with $\hat{p}$, a similar result is found where the final measurement is on $\hat{p}_3$. Our equations are linear in $\hat{q}$ and $\hat{p}$, meaning that any linear combination of the two, called a general quadrature, can be measured using Virtual Distillation. In the following numerical analysis, we apply this protocol to measure the quadratures of a noisy coherent state which endured dephasing.

Addressed improvements and continuations

We here derived exact formulas for the protocol in a very simple toy-model situation. In the following, we addressed the following extensions:

- Add noise to the protocol to ensure its robustness in a realistic setup,
- Start with a more accurate formula for the initial states than the toy model, based on what we expect experimentally,
- Try for other input states that have physically meaningful applications, such as cat state or GKP encoding.

4 Numerical analysis

In order to continue the analysis and assess the robustness of our protocol, different noise sources should be considered, and analytical computations become very quickly impossible. Numerical tools are thus needed, and we chose to use Python's library `qutip` to create a Jupyter notebook. We perform the whole protocol and then add several types of errors throughout the execution, mainly *incoherent* and *coherent* errors. The first type refers to decoherence, which causes mainly losses but also dephasing, and the second type occurs when the parameters of the parametrized gates $BS(\theta)$ are slightly off due to an imprecise drive of the control inputs.

We focused on error analysis and characterization for two main reasons. The first one is because our protocol aims to improve the performance in the case of a noisy input. We thus need a way to parametrize this "noisiness", in order to assess whether the protocol is indeed interesting or not, and in which regime. Secondly, we also need to numerically introduce errors during the protocol, to ensure that the protocol is robust and does not completely break down if some operations are not perfectly implemented.

4.1 Incoherent errors: losses

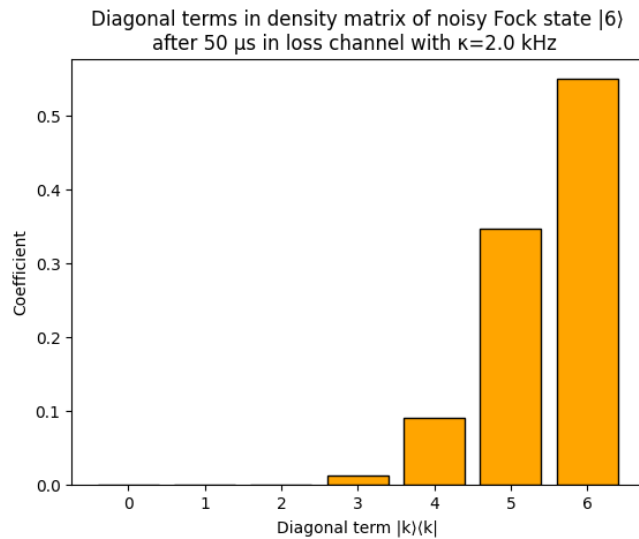


Figure 2: Example of noisy Fock state $|6\rangle$, represented with its diagonal coefficients in Fock basis

Incoherent errors happen because of interaction with the environment, which causes the system to decay over time and lose its coherence. When working with bosonic modes, we usually distinguish *losses* and *dephasing* as sources of decoherence. The first type has been described theoretically in section 2.2,

and the second type will be discussed below.

First of all, in order to test the protocol under various noise levels, we create several noisy Fock states of the form (4) for various values of t , which we report here for convenience.

$$\rho(t) = e^{-\kappa n t} \sum_{k=0}^n \binom{n}{k} (e^{\kappa t} - 1)^k |n-k\rangle\langle n-k|$$

This is a better form than Eq. (13) because it corresponds to a physical noise model, but it is in general harder to find closed forms for the relevant quantities.

In Figure 2, the diagonal coefficient terms in the density matrix of a noisy Fock state $|6\rangle$ are represented. We can see that mostly one photon losses have occurred from $|6\rangle$ to $|5\rangle$, but also two and even three losses with small probability.

In the first version of our procedure, we start by letting an ideal pure state $|\psi\rangle\langle\psi|$ evolve for some time t and enduring losses before starting the Virtual Distillation protocol. The x-axis in the plots corresponds to various times t ; points further to the right represent more and more noisy initial states. In order to have an estimate of the number operator, we then retrieve the expected values for the two operators $\hat{S}$ and $\hat{n}\hat{S}$ using Eq. (11) and (12) and compute their ratio. This quantity, which according to our mathematical analysis is equivalent to a measure of $\hat{n}$ on the distilled state, is represented along the y-axis. By increasing the initial time t , we expect the results to become worse (*i.e.* farther from the value of the pure state), but in comparison to the expected results *without* Virtual Distillation they should decrease more slowly. This is exactly what we see happening in Figure 3. When the noise is small, the protocol (orange curve) performs at its best, but even for very noisy inputs the protocol yields better results than a direct measurement of the $\hat{n}$ operator (red curve).

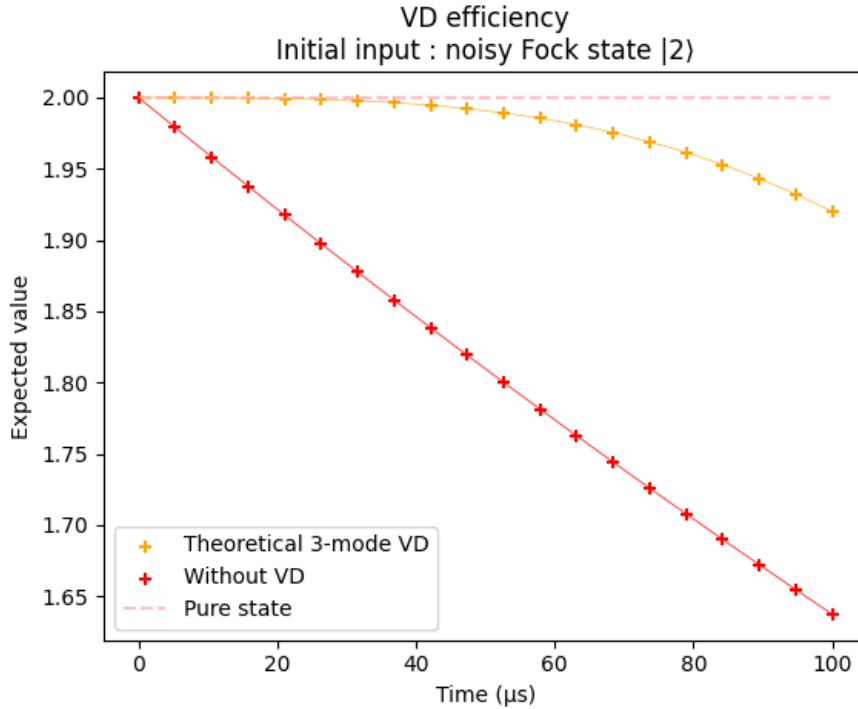


Figure 3: Theoretical results for the full Virtual Distillation protocol (orange curve) for the number operator $\hat{n}$ on initial Fock state $|2\rangle$, compared to direct measurement (red curve). The x-axis represents the amount of time the initial state has been exposed to a loss channel, its initial noisiness, and the y-axis represents the computed expected mean value of $\hat{n}$. Here the loss rate is $\kappa = \frac{1}{T_1}$ with $T_1 = 500\mu s$ the lifetime of a photon in the cavity.

Now that we have the theoretical plots, we can start adding some noise in the execution of the protocol itself. The first straightforward possibility is to add interleaved noise between each beam splitter, simulating waiting time between operations. However, as we found out in section 2.2, this interleaved loss commutes with the beam splitters and thus amounts to adding some noise at the beginning. This

effectively shifts the orange plot towards the left of a quantity τ representing the additional time the system has been exposed to the loss channel. Our numerical simulations show this exact behavior, visible in Figure 4.

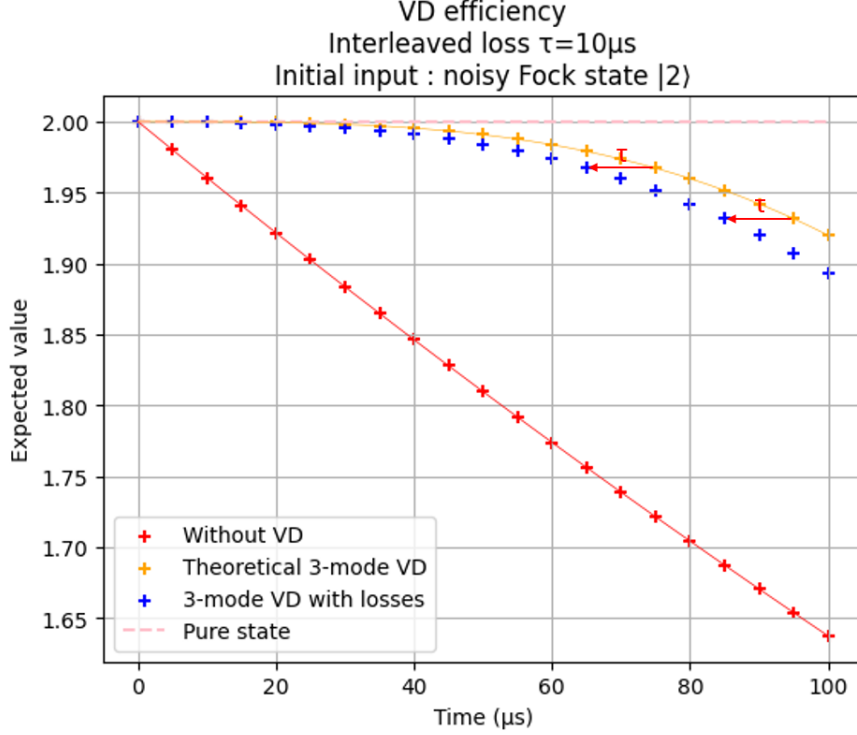


Figure 4: Full Virtual Distillation protocol, with theoretical results (orange) and results with interleaved loss errors (blue). For comparison the red curve shows the results when VD is not used. We can see that, as expected, the blue curve is simply shifted with respect to the orange one, by the amount corresponding to the interleaved losses added between the beam splitter operations.

4.2 Coherent errors

Another interesting kind of error is the over- or under-drive during the beam splitter operations. This would result in an angle θ not exactly equal to the desired angle, so the final $\hat{\mathcal{F}}_3$ gate will not be exactly the diagonalizing gate we seek. To model this kind of errors, we introduced some coherent noise in the angles following a Gaussian distribution

$$\varepsilon = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \sim \mathcal{N}(0, \sigma)$$

and an associated noisy $\hat{\mathcal{F}}_3$ gate generalizing Eq. (10):

$$\hat{\mathcal{F}}_3(\varepsilon) = e^{i(\frac{\pi}{4} + \varepsilon_1)(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1)} e^{i(\arcsin(\frac{1}{\sqrt{3}}) + \varepsilon_2)(\hat{a}_3^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_3)} e^{i(\frac{\pi}{4} + \varepsilon_3)(\hat{a}_3^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_3)}.$$

When adding this coherent noise, we observe in Figure 5 that the results remain noticeably better than without the protocol. We chose $\sigma = 10\% \times \frac{\pi}{4} \approx 0.08$, which is a large error considering the precision which can be reached in experiments.

As a remark, some values on the green plot exceed the pure state line, which may seem surprising as the total photon number should be below due to the loss channel. This is due to the fact that when $\hat{\mathcal{F}}_3$ is faulty, the computed quantity $\text{Tr}(\hat{n}\hat{S}) / \text{Tr}(\hat{S})$ is not equivalent to an expected value of the number operator anymore, and its range is thus not limited.

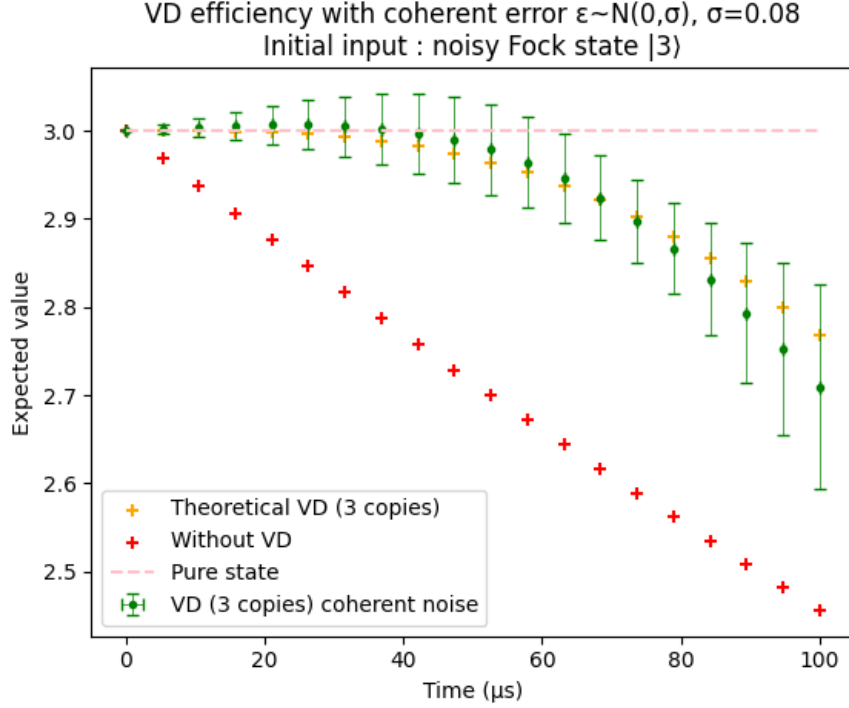


Figure 5: Theoretical and numerical results for full Virtual Distillation on Fock state $|3\rangle$ for $\hat{n}$ operator. During the execution, we add coherent noise and represent the average and standard deviation of the results over 30 noisy $\hat{\mathcal{T}}_3$ gates (green curve with the error bars).

By combining coherent and interleaved incoherent errors, we obtained as in Figure 4 the same plot as the green one but shifted to the left.

4.3 Other input states

The protocol works well for input Fock states, because the form of the density matrix for a noisy Fock state after losses is very well suited for this kind of protocol. Indeed, the dominant eigenvector of ρ remains the pure state that we want to distill. When trying to extend the protocol for other input states, it is not obvious that the Virtual Distillation method should continue to work. In [18], a theoretical work shows that when the target state is an eigenstate of the measured observable, then the coherent drift of the main eigenvector caused by a general noise channel is exponentially less severe than the incoherent loss of fidelity, which is corrected by Virtual Distillation. In our case, this justifies that for Fock input state and number operator measurement, VD yields very good results. However, for other input states, the regime is not the same anymore and we should not expect very good results.

Using the working script, we reproduced the results for different input states, in order to numerically assess the performances.

Binomial state

We start from the binomial state $|\psi\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}}$, and then perform the whole protocol.

As seen from Figure 6, the yellow line lies under the red one, meaning that distillation is not working and the protocol is not fit for binomial state when measuring the number operator. We can understand this through the fact that a pure loss channel causes a huge drift in the dominant eigenvector of the binomial state (because of the asymmetry between $|0\rangle$ and $|4\rangle$). If we perform a full analytical analysis

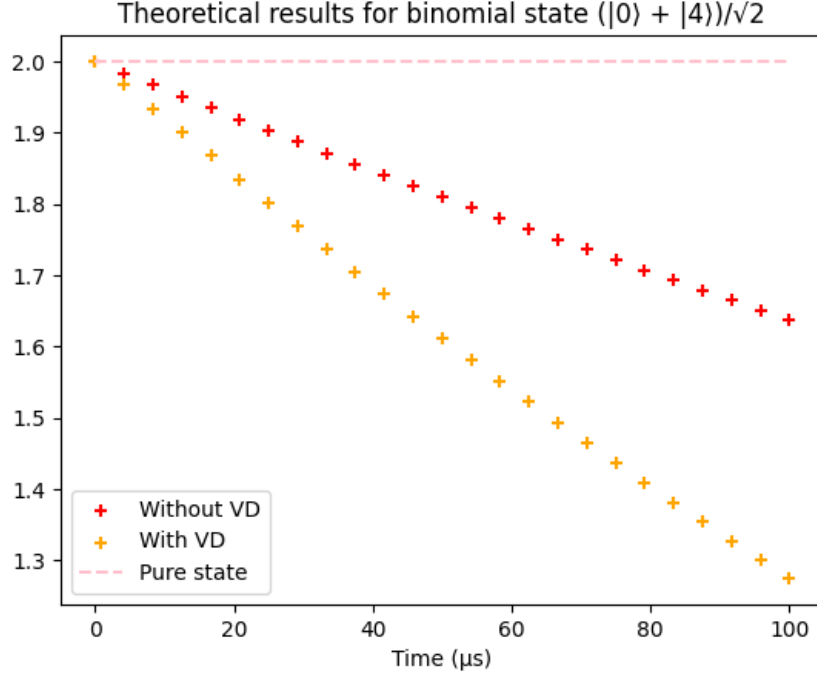


Figure 6: Theoretical plots for initial Binomial state, with $\kappa = \frac{1}{T_1}$ and $T_1 = 500\mu s$. Even theoretically, the protocol performs worse than when doing nothing.

of the scenario starting from $\rho_0 = \frac{1}{2}(|0\rangle + |4\rangle)(\langle 0| + \langle 4|)$ and putting it in a loss channel with strength $\gamma = \kappa t$, we see that the main terms are

$$\rho(\gamma) = \frac{1}{2} ((|0\rangle + e^{-2\gamma}|4\rangle)(\langle 0| + e^{-2\gamma}\langle 4|) + 4(e^\gamma - 1)e^{-4\gamma}|3\rangle\langle 3| + \dots).$$

The main eigenvector $\frac{1}{\sqrt{2}}(|0\rangle + |4\rangle)$ has drifted to $\frac{1}{\sqrt{1+e^{-4\gamma}}}(|0\rangle + e^{-2\gamma}|4\rangle)$, and the VD protocol gets rid of the contribution of $|3\rangle\langle 3|$, thus lowering the average photon number with respect to a direct measurement.

This led us to consider more general versions of the binomial state of the form $\frac{|i\rangle + |j\rangle}{\sqrt{2}}$. The analytical analysis becomes more tedious as the density matrix is not easily diagonalizable, but we observed numerically that Virtual Distillation works when $i > 0$ and $j < 3i$ (empirical observation), as we can see on the plot in Figure 7.

Cat state

Cat states are a class of states that allow for encoding logical qubits into bosonic modes. Their usefulness has been extensively studied, so it can be worth trying to see if Virtual Distillation works for such target states. We define one of the cat states as follows:

$$|\text{cat}\rangle = \frac{|\alpha\rangle + |-\alpha\rangle}{\sqrt{2}}$$

(for $|\alpha|$ large enough (typically > 1) otherwise the normalization is different).

In Figure 8 we see that the protocol does not yield a better result directly from the theoretical aspect, and coherent errors make the results deviate from the theoretical ones. As we have seen, coherent states are not suited for Virtual Distillation, and cat states inherit essentially a lot of properties from coherent states. This could explain why the protocol gives worse results than doing nothing for the case of target cat states.

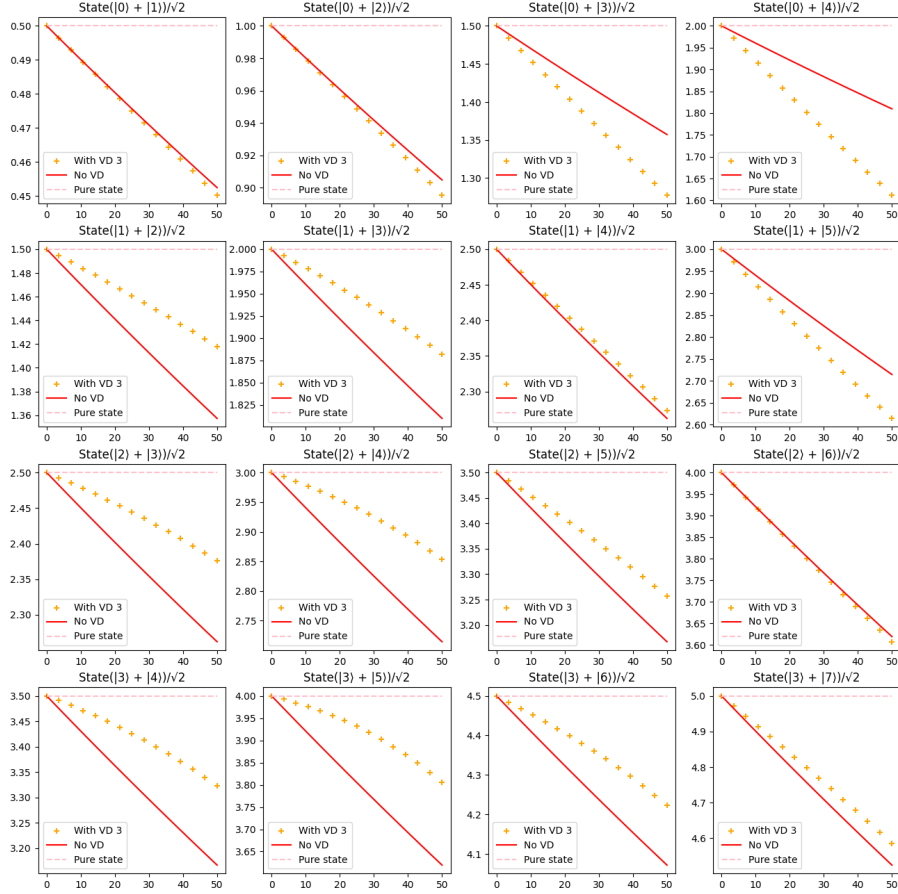


Figure 7: Comparison of theoretical results for measurements of number operator using Virtual Distillation on various generalized binomial states. We observe that apart from the top row and right-most column, the orange points are above the red plot, meaning that Virtual Distillation is indeed yielding better results.

GKP state

The GKP state [21] is written as an (unnormalized) sum of eigenstates of position operator $\hat{q}$:

$$|0\rangle_{\text{GKP}}^{(\text{ideal})} = \sum_{n=-\infty}^{\infty} |2n\sqrt{\pi}\rangle_q.$$

However, in reality we often consider a finitely-squeezed version, parametrized by some δ :

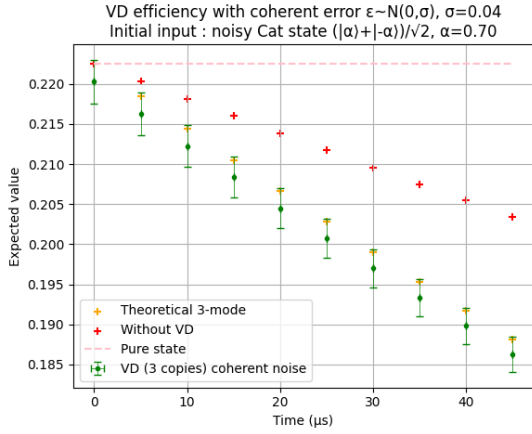
$$|0_\delta\rangle = e^{-\delta^2(\hat{n} + \frac{1}{2})} |0\rangle_{\text{GKP}}^{(\text{ideal})}.$$

It turns out that the state can be written in a way easier to implement ([22, 23]) as:

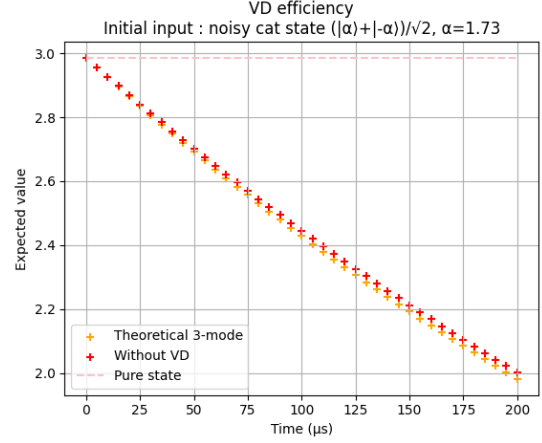
$$|0_\delta\rangle \propto \sum_{s \in \mathbb{Z}} e^{-\pi s^2 \tanh \delta^2} \hat{D}(2s\sqrt{\pi}) \hat{S}(-\ln \sqrt{\sinh \delta^2 \cosh \delta^2}) |0\rangle,$$

with $\hat{D}$ and $\hat{S}$ respectively squeezing and displacement operator.

For similar reasons as for the binomial state, we see from Figure 9 that the protocol is not suited for GKP states either, as the dominant eigenvector endures a drift under our loss model. Our numerical results (the green curve not plotted here) turn out to be completely off due to errors stemming from the truncation of Hilbert space. In fact, finitely squeezed GKP states have non-zero coefficients up to high photon numbers, and our simulations could not reach this regime as it would require too much memory and time ($N^3 \times N^3$ matrices when truncating at N). The standard `qutip` library does not seem suited for such computations, thus limiting the simulations we can perform.



(a) Plot with the noisy results averaged on 20 faulty gates. We chose here $\alpha = 0.7$.



(b) Plot with only the theoretical curves for a higher value of $\alpha = 1.73$.

Figure 8: Virtual Distillation for cat state, $T_1 = 500\mu s$.

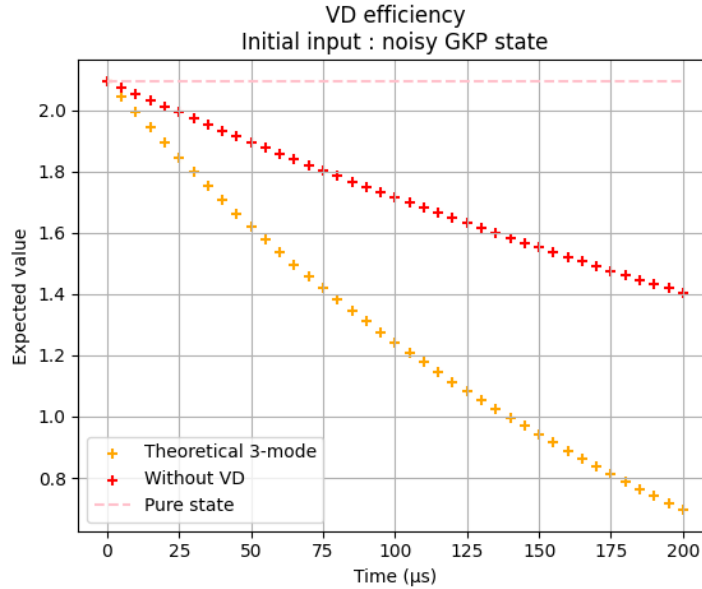


Figure 9: Virtual Distillation for GKP state. The yellow line lies under the red one, meaning that the protocol is not working. The numerical results are not relevant due to Hilbert space truncation errors.

4.4 Dephasing errors

For now, we only focused on losses as sources of incoherent noise. Another kind of error that may occur when dealing with bosonic modes, typically in an optical microwave cavity, are *dephasing errors*. Like losses, they can be described through the Lindblad formalism using Kraus operators (see [17] appendix F for details):

$$\hat{B}_\ell(t) = \frac{(\kappa_\phi t)^{\ell/2}}{\sqrt{\ell!}} e^{-\kappa_\phi \hat{n}^2 t/2} \hat{n}^\ell$$

$$\rho(t) = e^{t\mathcal{L}_{\kappa_\phi}} \rho(0) = \sum_{\ell=0}^{\infty} \hat{B}_\ell(t) \rho(0) \hat{B}_\ell(t)^\dagger$$

In the following plot, we performed a similar calculation as before, but now the input state is a coherent state $|\alpha\rangle$ enduring *dephasing errors* before entering the protocol. We then measure its $\hat{q}$ quadrature using the second version of the protocol described in section 3.2.3.

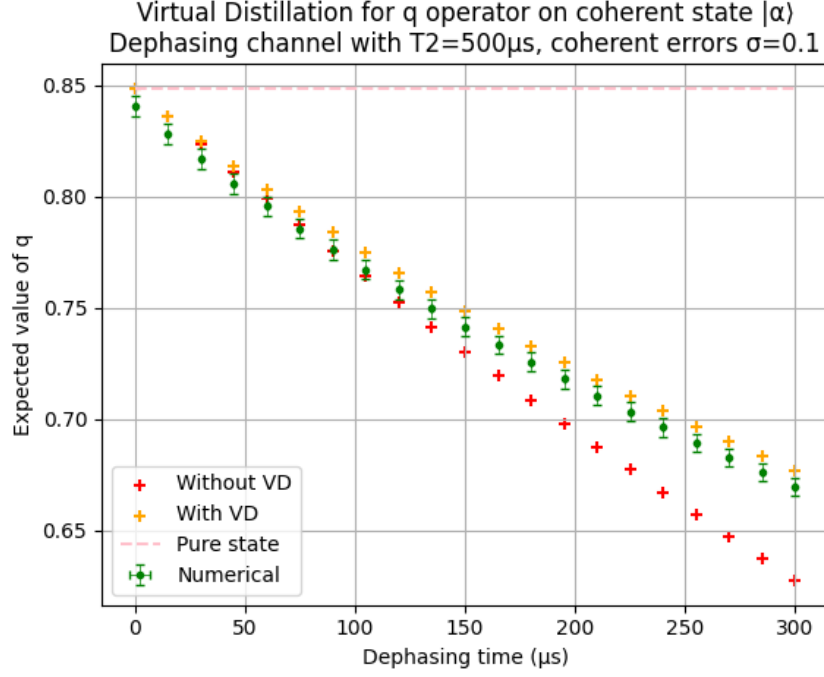


Figure 10: Quadrature measurement using Virtual Distillation on dephased (noisy) coherent state $|\alpha\rangle$. Here we chose $\alpha = 0.6 + 0.3i$, and a dephasing channel with lifetime $T_2 = 500\mu s$. Numerical results take into account coherent errors in the beam splitter operations.

Even if Virtual Distillation for quadrature operator on coherent state is not as impressive as for number operator on Fock states, it still brings some benefit with respect to a direct measurement. Moreover, on Figure 11 we observe that the protocol performs really well on dephased squeezed coherent state of the form

$$\hat{S}(\xi) |\alpha\rangle \quad \text{with} \quad \hat{S}(\xi) = e^{i\xi(\hat{p}\hat{q} + \hat{q}\hat{p})/2}$$

when $\xi = 0.5$ (squeezing in the q direction) and performs well if noise is above some threshold for $\xi = -0.5$ (squeezing in the p direction). A general description of which initial input states are suited for the protocol has not been achieved yet, and these results could be better understood by a more complete analysis of the dominant eigenvector coherent drift. What we observe nonetheless is that when the initial input is close to an eigenstate of the observable (which would be an infinitely q -squeezed state), then the protocol works consistently, which was the conclusion drawn from [18] on the theoretical level.

Lastly, we tried to numerically assess all combinations of the concepts introduced above, which we can summarize in the table of Figure 12.

Wigner functions and VD quadrature measurements on squeezed coherent state.
Dephasing $T_2=500\mu s$

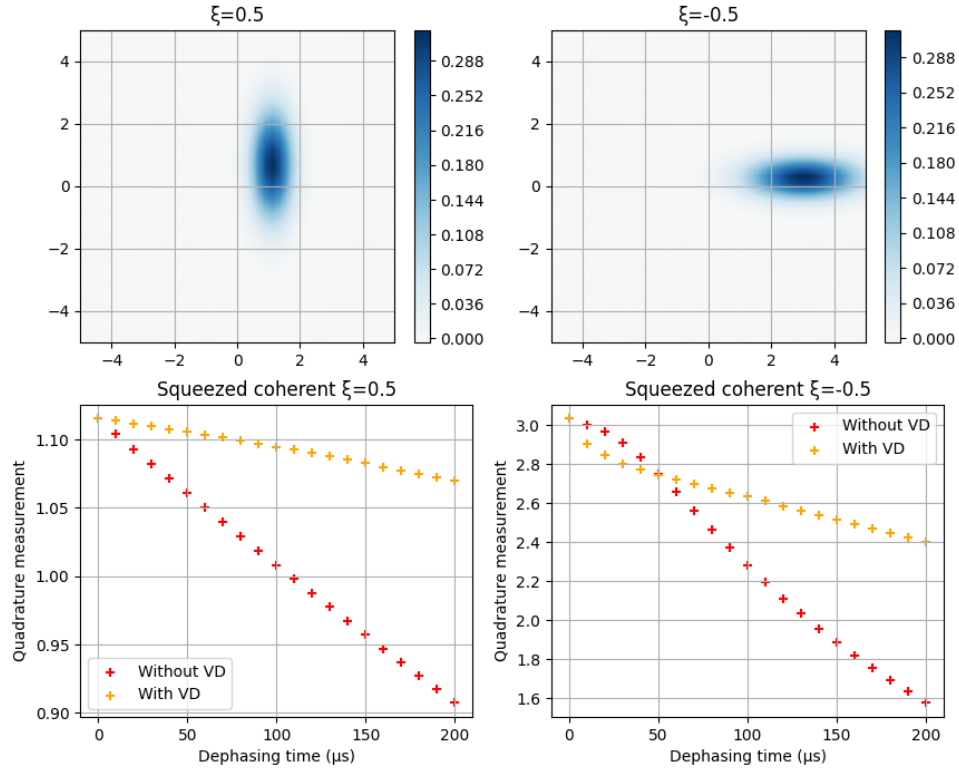


Figure 11: Theoretical results for Virtual Distillation on squeezed coherent states. The plots reproduce the results with a dephased input for quadrature measurement. We chose $\alpha = 1.3 + 0.6i$, $\xi = \pm 0.5$ and $T_2 = 500\mu s$.

Measured observable	$\hat{n}$		$\hat{q}$	
Type of initial noise	Losses	Dephasing	Losses	Dephasing
Fock	Yes	-	0	
Coherent	No	No	No	Yes
Binomial	In some cases	No	0	
Cat state	No	No	0	
GKP	No	From some noise level	0	
Squeezed coherent	No	No	No	Yes

Figure 12: Summary of tested regimes for Virtual Distillation with working and non-working results. "-": the state is not affected by the noise channel, "0": expected value is always zero.

5 Conclusion

Virtual Distillation is a promising tool for Quantum Error Mitigation techniques, as it allows to reach better approximations of the mean value of a given observable, using only a few copies of a given state. To be used in practice, it requires an increase in the number of operations, but much less than what would be needed for full quantum error correction. Measuring the mean value of an observable is common in quantum algorithms, and if we know that the state before measurement has a dominant eigenvector close to an eigenstate of the observable, then VD is proven to bring real benefits. For bosonic modes, we found an explicit construction for a quantum circuit performing the protocol for $M = 3$ copies, which was not the case when dealing with qubits (reference [13] only mentions an approximate circuit for 3-copies qubit VD).

Our numerical analysis ensured the robustness of the protocol to interleaved losses and dephasing, as well as coherent errors in the drive of the beam splitter operations. The focus was put on the best-working regime where we look for the mean value of $\hat{n}$ on a noisy Fock state $|n\rangle$, and we confirmed both theoretically and numerically the benefit of the protocol. Using a very similar circuit, we achieved the same results for a measurement of the $\hat{q}$ quadrature on dephased coherent states. However, not all states are suited for Virtual Distillation, and in our analysis it turned out that for measurements of the number operator on other input states than Fock states, Virtual Distillation could rarely bring an advantage, mostly due to a coherent drift in the dominant eigenvector of the state under our loss models.

In future work, we could consider circuits with more modes, for example a 4-mode setup which could possibly be realized using beam splitters and phase shifters according to [19]. We think that it is possible to scale for $M \geq 5$ using results from [20], but numerical simulations would quickly become intractable. It would also be interesting to look at other observables, and define categories of states and measurements for which VD brings a benefit by better understanding how the coherent eigenvector drift behaves. In our approach, in order to perform general measurements we would require a new diagonalizing gate expressing $\hat{S}$ and $\hat{O}\hat{S}$ in terms of observables that we can measure in practice (*e.g.* parity, homodyne, heterodyne).

Additionally, similar techniques as those developed for qubit Virtual Distillation could be transposed to bosonic systems, such as measurements on auxiliary qubits as developed in [13] and assessed in [24]. For our case of bosonic modes, this would require a hybrid C-SWAP gate with a qubit control, which has not been achieved yet at Chalmers.

Bibliography

- [1] Alpha Gaëtan et al. “Observation of collective excitation of two individual atoms in the Rydberg blockade regime”. In: *Nature Physics* 5.2 (Jan. 2009), pp. 115–118. ISSN: 1745-2481. DOI: 10.1038/nphys1183.
- [2] Seth Lloyd. “Universal Quantum Simulators”. In: *Science* 273.5278 (1996), pp. 1073–1078. DOI: 10.1126/science.273.5278.1073.
- [3] Peter W. Shor. “Algorithms for quantum computation: Discrete logarithms and factoring”. In: *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*. IEEE. 1994, pp. 124–134.
- [4] Peter W. Shor. “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM Journal on Computing* 26.5 (Oct. 1997), pp. 1484–1509. ISSN: 1095-7111. DOI: 10.1137/S0097539795293172. URL: <http://dx.doi.org/10.1137/S0097539795293172>.
- [5] C.S. Adams and E. Riis. “Laser cooling and trapping of neutral atoms”. In: *Progress in Quantum Electronics* 21.1 (1997), pp. 1–79. ISSN: 0079-6727. DOI: [https://doi.org/10.1016/S0079-6727\(96\)00006-7](https://doi.org/10.1016/S0079-6727(96)00006-7). URL: <https://www.sciencedirect.com/science/article/pii/S0079672796000067>.
- [6] P. Krantz et al. “A quantum engineer’s guide to superconducting qubits”. In: *Applied Physics Reviews* 6.2 (June 2019). ISSN: 1931-9401. DOI: 10.1063/1.5089550. URL: <http://dx.doi.org/10.1063/1.5089550>.
- [7] John Preskill. “Quantum Computing in the NISQ era and beyond”. In: *Quantum* 2 (Aug. 2018), p. 79. ISSN: 2521-327X. DOI: 10.22331/q-2018-08-06-79. URL: <https://doi.org/10.22331/q-2018-08-06-79>.
- [8] Austin G. Fowler et al. “Surface codes: Towards practical large-scale quantum computation”. In: *Phys. Rev. A* 86 (3 Sept. 2012), p. 032324. DOI: 10.1103/PhysRevA.86.032324. URL: <https://link.aps.org/doi/10.1103/PhysRevA.86.032324>.
- [9] A. M. Steane. “Error Correcting Codes in Quantum Theory”. In: *Phys. Rev. Lett.* 77 (5 July 1996), pp. 793–797. DOI: 10.1103/PhysRevLett.77.793. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.77.793>.
- [10] Abhinav Kandala et al. “Error mitigation extends the computational reach of a noisy quantum processor”. In: *Nature* 567 (Mar. 2019). ISSN: 1476-4687. DOI: 10.1038/s41586-019-1040-7.
- [11] Zhenyu Cai et al. “Quantum error mitigation”. In: *Reviews of Modern Physics* 95.4 (Dec. 2023). ISSN: 1539-0756. DOI: 10.1103/revmodphys.95.045005. URL: <http://dx.doi.org/10.1103/RevModPhys.95.045005>.
- [12] Suguru Endo et al. *Quantum error mitigation for rotation symmetric bosonic codes with symmetry expansion*. 2022. arXiv: 2211.06164 [quant-ph]. URL: <https://arxiv.org/abs/2211.06164>.
- [13] William J. Huggins et al. “Virtual Distillation for Quantum Error Mitigation”. In: *Physical Review X* 11.4 (Nov. 2021). ISSN: 2160-3308. DOI: 10.1103/physrevx.11.041036.
- [14] Bálint Koczor. “Exponential Error Suppression for Near-Term Quantum Devices”. In: *Physical Review X* 11.3 (Sept. 2021). ISSN: 2160-3308. DOI: 10.1103/physrevx.11.031057. URL: <http://dx.doi.org/10.1103/PhysRevX.11.031057>.

- [15] T.E. O'Brien et al. "Purification-based quantum error mitigation of pair-correlated electron simulations". In: *Nat. Phys.* 19 (2023), pp. 1782–1792. DOI: <https://doi.org/10.1038/s41567-023-02240-y>.
- [16] Jordan Cotler et al. "Quantum Virtual Cooling". In: *Physical Review X* 9.3 (July 2019). ISSN: 2160-3308. DOI: 10.1103/physrevx.9.031013.
- [17] Arne L. Grimsmo, Joshua Combes, and Ben Q. Baragiola. "Quantum Computing with Rotation-Symmetric Bosonic Codes". In: *Physical Review X* 10.1 (Mar. 2020). ISSN: 2160-3308. DOI: 10.1103/physrevx.10.011058. URL: <http://dx.doi.org/10.1103/PhysRevX.10.011058>.
- [18] Bálint Koczor. "The dominant eigenvector of a noisy quantum state". In: *New Journal of Physics* 23.12 (Dec. 2021), p. 123047. ISSN: 1367-2630. DOI: 10.1088/1367-2630/ac37ae.
- [19] Marek Żukowski, Anton Zeilinger, and Michael A. Horne. "Realizable higher-dimensional two-particle entanglements via multiport beam splitters". In: *Phys. Rev. A* 55 (4 Apr. 1997), pp. 2564–2579. DOI: 10.1103/PhysRevA.55.2564. URL: <https://link.aps.org/doi/10.1103/PhysRevA.55.2564>.
- [20] Igor Jex, Stig Stenholm, and Anton Zeilinger. "Hamiltonian theory of a symmetric multiport". In: *Optics Communications* 117.1 (1995), pp. 95–101. ISSN: 0030-4018. DOI: [https://doi.org/10.1016/0030-4018\(95\)00078-M](https://doi.org/10.1016/0030-4018(95)00078-M). URL: <https://www.sciencedirect.com/science/article/pii/003040189500078M>.
- [21] Ilan Tzitrin et al. "Progress towards practical qubit computation using approximate Gottesman-Kitaev-Preskill codes". In: *Physical Review A* 101.3 (Mar. 2020). ISSN: 2469-9934. DOI: 10.1103/physreva.101.032315. URL: <http://dx.doi.org/10.1103/PhysRevA.101.032315>.
- [22] Takaya Matsuura, Hayata Yamasaki, and Masato Koashi. "Equivalence of approximate Gottesman-Kitaev-Preskill codes". In: *Physical Review A* 102.3 (Sept. 2020). ISSN: 2469-9934. DOI: 10.1103/physreva.102.032408. URL: <http://dx.doi.org/10.1103/PhysRevA.102.032408>.
- [23] Timo Hillmann et al. "Performance of Teleportation-Based Error-Correction Circuits for Bosonic Codes with Noisy Measurements". In: *PRX Quantum* 3.2 (May 2022). ISSN: 2691-3399. DOI: 10.1103/prxquantum.3.020334. URL: <http://dx.doi.org/10.1103/PRXQuantum.3.020334>.
- [24] Pontus Vikstål, Giulia Ferrini, and Shruti Puri. *Study of noise in virtual distillation circuits for quantum error mitigation*. 2024. arXiv: 2210.15317 [quant-ph]. URL: <https://arxiv.org/abs/2210.15317>.

Appendices

A Properties of a loss channel

General properties of a loss channel

Loss channels are a central part of our analysis, and it is worth looking a bit more in detail to its main characteristics. First of all, let's check that the resulting density operator is valid, *i.e.* that its trace is 1. For simplicity we write $\gamma = \kappa t$.

$$\begin{aligned}\text{Tr}(\rho(t)) &= \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} \text{Tr} \left(e^{-\gamma \hat{n}/2} \hat{a}^k \rho_0 \hat{a}^{\dagger k} e^{-\gamma \hat{n}/2} \right) \\ &= \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} \text{Tr} (\rho_0 \hat{a}^{\dagger k} e^{-\gamma \hat{n}} \hat{a}^k) \quad \text{by cyclicity of trace} \\ &= \text{Tr} \left(\rho_0 \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} \hat{a}^{\dagger k} \hat{a}^k e^{-\gamma \hat{n}} \right),\end{aligned}$$

where we used the general fact that for any analytical function f ,

$$f(\hat{a}^{\dagger} \hat{a} + 1) \hat{a} = f(\hat{a} \hat{a}^{\dagger}) \hat{a} = (c_0 + c_1 \hat{a} \hat{a}^{\dagger} + c_2 \hat{a} \hat{a}^{\dagger} \hat{a} \hat{a}^{\dagger} + \dots) \hat{a} = \hat{a} (c_0 + c_1 \hat{a}^{\dagger} \hat{a} + c_2 \hat{a}^{\dagger} \hat{a} \hat{a}^{\dagger} \hat{a} + \dots) = \hat{a} f(\hat{a}^{\dagger} \hat{a})$$

so applying for $f(x) = e^{-\gamma(x-1)/2}$, we obtain that

$$e^{-\gamma \hat{a}^{\dagger} \hat{a}} \hat{a} = \hat{a} e^{\gamma} e^{-\gamma \hat{a}^{\dagger} \hat{a}} \quad \text{and thus} \quad e^{-\gamma \hat{n}} \hat{a}^k = \hat{a}^k e^{-\gamma \hat{n}} e^{\gamma k}.$$

We then use the fact that $\hat{a}^{\dagger k} \hat{a}^k = \hat{n}(\hat{n} - 1) \dots (\hat{n} - k + 1) = \frac{\hat{n}!}{(\hat{n} - k)!}$ to write more compactly that

$$\begin{aligned}\text{Tr}(\rho(t)) &= \text{Tr} \left(\rho_0 \sum_{k=0}^{\infty} (e^{\gamma} - 1)^k \binom{\hat{n}}{k} e^{-\gamma \hat{n}} \right) \\ &= \text{Tr} (\rho_0 (e^{\gamma} - 1 + 1)^{\hat{n}} e^{-\gamma \hat{n}}) = \text{Tr}(\rho_0) = 1.\end{aligned}$$

A loss channel is mainly used to represent losses accumulating over time inside a cavity (due to imperfect mirrors for example). We can formalize this idea by calculating the evolution of the average photon number in the mode:

$$\begin{aligned}
\langle \hat{n} \rangle(t) &= \text{Tr}(\hat{n}\rho(t)) = \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} \text{Tr}(\rho_0 \hat{a}^{\dagger k} \hat{n} e^{-\gamma \hat{n}} \hat{a}^k) \\
&= \sum_{k=0}^{\infty} \frac{(1 - e^{-\gamma})^k}{k!} \text{Tr}(\rho_0 \hat{a}^{\dagger k} \hat{a}^k (\hat{n} - k) e^{-\gamma(\hat{n} - k)}) \\
&= \text{Tr} \left(\rho_0 \sum_{k=0}^{\infty} (e^{\gamma} - 1)^k \binom{\hat{n}}{k} (\hat{n} - k) e^{-\gamma \hat{n}} \right) \quad \text{we can use that } (n - k) \binom{n}{k} = n \binom{n-1}{k} \\
&= \text{Tr}(\rho_0 \hat{n} (e^{\gamma} - 1 + 1)^{\hat{n}-1} e^{-\gamma \hat{n}}) = e^{-\gamma} \text{Tr}(\rho_0 \hat{n}) \\
\langle \hat{n} \rangle(t) &= e^{-\kappa t} \langle \hat{n} \rangle(0).
\end{aligned}$$

We see that the average photon number follows an exponential decay, with a decay rate κ , as expected by the intuitive interpretation.

Loss channel on coherent state

Let us start with the effect of a loss channel on a coherent state $|\alpha\rangle$, with associated density matrix $\rho_0 = |\alpha\rangle\langle\alpha|$. We can start by looking at the action of one of the Kraus operators:

$$\begin{aligned}
\hat{A}_k |\alpha\rangle &= \frac{\overbrace{(1 - e^{-\kappa t})^{k/2}}^{c_k(t)}}{\sqrt{k!}} e^{-\kappa \hat{n} t/2} \hat{a}^k |\alpha\rangle \\
&= c_k(t) e^{-\kappa \hat{n} t/2} \alpha^k |\alpha\rangle \\
&= c_k(t) \alpha^k e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-\kappa \hat{n} t/2} |n\rangle \\
&= c_k(t) \alpha^k e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha e^{-\kappa t/2})^n}{\sqrt{n!}} |n\rangle \\
&= c_k(t) \alpha^k e^{-\frac{|\alpha|^2}{2}(1 - e^{-\kappa t})} |\alpha e^{-\kappa t/2}\rangle.
\end{aligned}$$

The dependence in k is well-localized, so we can compute the full state using Eq. (2)

$$\rho(t) = \underbrace{\left(\sum_{k=0}^{\infty} c_k(t)^2 |\alpha|^{2k} \right)}_{\exp((1 - e^{-\kappa t})|\alpha|^2)} e^{-|\alpha|^2(1 - e^{-\kappa t})} |\alpha e^{-\kappa t/2}\rangle\langle\alpha e^{-\kappa t/2}| = |\alpha e^{-\kappa t/2}\rangle\langle\alpha e^{-\kappa t/2}|.$$

A pure loss channel has thus the effect of exponentially damping a coherent state, and preserving its purity. This excludes a coherent state from the possible candidates of the Virtual Distillation protocol.

Loss channel on Fock state

Let's now focus on a different example, Fock states:

$$\begin{aligned}
\hat{A}_k |n\rangle &= c_k(t) e^{-\kappa \hat{n} t/2} \hat{a}^k |n\rangle \\
&= c_k(t) e^{-\kappa(n-k)t/2} \sqrt{\frac{n!}{(n-k)!}} |n-k\rangle \\
&= (1 - e^{-\kappa t})^{k/2} e^{-\kappa(n-k)t/2} \sqrt{\binom{n}{k}} |n-k\rangle \\
&= (e^{\kappa t} - 1)^{k/2} e^{-\kappa n t/2} \sqrt{\binom{n}{k}} |n-k\rangle.
\end{aligned}$$

And putting everything in (2), we obtain that

$$\rho(t) = e^{-\kappa n t} \sum_{k=0}^n \binom{n}{k} (e^{\kappa t} - 1)^k |n-k\rangle\langle n-k|. \quad (14)$$

This expression is exactly of the form of Eq. (6), thus making Fock states well suited for our VD protocol.

B Proof of diagonalizing formula

We will here prove the formula (9). This formula has been introduced in [16] but without proof, and the following proof is original. In the article, the letter n was used instead of M , and here we follow the same convention. We recall the full statement:

Proposition. Define the unitary $\hat{\mathcal{F}}_n$ called "discrete Fourier transform" such that

$$\hat{\mathcal{F}}_n^\dagger \hat{a}_p \hat{\mathcal{F}}_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n e^{2i\pi \frac{kp}{n}} \hat{a}_k \quad \text{for } p \in \{1, \dots, n\}.$$

If we consider the operator $\hat{\mathcal{R}}_n$, defined as

$$\hat{\mathcal{R}}_n = \exp \left\{ 2i\pi \frac{1}{n} \sum_{p=1}^n p \hat{n}_p \right\},$$

then we have the following identity:

$$\hat{\mathcal{F}}_n^\dagger \hat{\mathcal{R}}_n \hat{\mathcal{F}}_n = \hat{S}_n.$$

For the proof, we rely on the following result:

Lemma 1. Let J be a Hermitian $n \times n$ matrix, representing a generic purely quadratic Hamiltonian $\hat{\mathcal{H}} = J_{kl} \hat{a}_k^\dagger \hat{a}_l$, and the associated unitary $\hat{U} = e^{i\hat{\mathcal{H}}}$. Then, the action of $\hat{U}$ on the ladder operators is given by

$$\hat{U}^\dagger \hat{a}_m \hat{U} = \exp\{iJ\}_{ml} \hat{a}_l,$$

where $\exp$ is the matrix exponentiation.

Proof. Let's define $\hat{U}(\theta) = \exp\{i\theta J_{kl} \hat{a}_k^\dagger \hat{a}_l\}$, so that $\hat{U} = \hat{U}(1)$, and write $\hat{A}_m(\theta) = \hat{U}^\dagger(\theta) \hat{a}_m \hat{U}(\theta)$. Note that $\hat{A}_m(0) = \hat{a}_m$. We have

$$\begin{aligned} \frac{d}{d\theta} \hat{A}_m(\theta) &= \frac{d}{d\theta} \left(\hat{U}^\dagger(\theta) \hat{a}_m \hat{U}(\theta) \right) \\ &= \left(\frac{d}{d\theta} \exp\{-i\theta J_{kl} \hat{a}_k^\dagger \hat{a}_l\} \right) \hat{a}_m \hat{U}(\theta) + \hat{U}^\dagger(\theta) \hat{a}_m \left(\frac{d}{d\theta} \exp\{i\theta J_{kl} \hat{a}_k^\dagger \hat{a}_l\} \right) \\ &= \hat{U}^\dagger(\theta) \left(-iJ_{kl} \hat{a}_k^\dagger \hat{a}_l \right) \hat{a}_m \hat{U}(\theta) + \hat{U}^\dagger(\theta) \hat{a}_m \left(iJ_{kl} \hat{a}_k^\dagger \hat{a}_l \right) \hat{U}(\theta) \\ &= iJ_{kl} \hat{U}^\dagger(\theta) \underbrace{\left[\hat{a}_m, \hat{a}_k^\dagger \hat{a}_l \right]}_{\delta_{mk} \hat{a}_l} \hat{U}(\theta) \\ &= iJ_{ml} \hat{A}_l(\theta). \end{aligned}$$

This set of linear differential equations can be compactly solved using matrix exponentiation notation:

$$\hat{A}_m(\theta) = \exp\{i\theta J\}_{ml} \hat{A}_l(0).$$

By setting $\theta = 1$ we then prove the lemma. □

Let's now focus on the operator $\hat{\mathcal{F}}_n^\dagger \hat{\mathcal{R}}_n \hat{\mathcal{F}}_n$. We wish to prove that its action on the $\hat{a}_p$ is the same as the cyclic shift $\hat{S}_n$. We have

$$\begin{aligned} \hat{\mathcal{F}}_n^\dagger \hat{\mathcal{R}}_n \hat{\mathcal{F}}_n &= \hat{\mathcal{F}}_n^\dagger \exp \left\{ 2i\pi \frac{1}{n} \sum_{p=1}^n p \hat{n}_p \right\} \hat{\mathcal{F}}_n \\ &= \exp \left\{ 2i\pi \frac{1}{n} \sum_{p=1}^n p \underbrace{\left(\hat{\mathcal{F}}_n^\dagger \hat{n}_p \hat{\mathcal{F}}_n \right)}_{(\hat{\mathcal{F}}_n^\dagger \hat{a}_p^\dagger \hat{\mathcal{F}}_n)(\hat{\mathcal{F}}_n^\dagger \hat{a}_p \hat{\mathcal{F}}_n)} \right\} \\ &= \exp \left\{ 2i\pi \frac{1}{n} \sum_{p=1}^n p \cdot \frac{1}{n} \sum_{k,l=1}^n e^{-2i\pi \frac{kp}{n}} e^{2i\pi \frac{lp}{n}} \hat{a}_k^\dagger \hat{a}_l \right\}, \end{aligned}$$

where in the last step we used the definition of the discrete Fourier transform operator. We can then write

$$\hat{\mathcal{F}}_n^\dagger \hat{\mathcal{R}}_n \hat{\mathcal{F}}_n = \exp \left\{ i \hat{a}_k^\dagger J_{kl} \hat{a}_l \right\}$$

where, by introducing $\omega = e^{2i\pi/n}$,

$$J_{kl} = \frac{2\pi}{n^2} \sum_{p=1}^n p \omega^{p(l-k)}.$$

By using the formula $\sum_{p=1}^n p \alpha^p = \frac{\alpha (n\alpha^{n+1} - (n+1)\alpha^n + 1)}{(\alpha - 1)^2}$ for $\alpha \neq 1$ (obtained by differentiating the standard geometric summation formula), we can simplify

$$\begin{aligned} \text{if } l \neq k, \quad J_{kl} &= \frac{2\pi}{n^2} \frac{\omega^{l-k} (n\omega^{l-k} - (n+1) + 1)}{(\omega^{l-k} - 1)^2} \\ &= \frac{2\pi}{n} \frac{\omega^{l-k}}{\omega^{l-k} - 1}, \\ \text{if } l = k, \quad J_{kl} &= \frac{2\pi}{n} \frac{n+1}{2}. \end{aligned}$$

According to the lemma we stated earlier, we know wish to compute $\exp\{iJ\}$ to know the action of the operator on the $\hat{a}_p$. Hopefully this would yield the well known cyclic permutation matrix.

Upon close inspection, it appears that the matrix J is **circulant**, *i.e.* constant along its diagonals (J_{kl} only depends on $l-k$). We know that circulant matrices have eigenvectors $v_j = (1, \omega^j, \omega^{2j}, \dots, \omega^{(n-1)j})^T$ with associated eigenvalues

$$\lambda_j = J_{1m}(v_j)_m = \frac{2\pi}{n} \left(\frac{n+1}{2} + \frac{\omega^{j+1}}{\omega - 1} + \frac{\omega^{2(j+1)}}{\omega^2 - 1} + \dots + \frac{\omega^{(n-1)(j+1)}}{\omega^{n-1} - 1} \right).$$

Let's consider the diagonalizing matrix $\Omega = (\omega^{(k-1)(l-1)})_{k,l=1,\dots,n}$, such that

$$J = \Omega \text{Diag}(\lambda_0, \dots, \lambda_{n-1}) \Omega^{-1} \quad (15)$$

$$\exp\{iJ\} = \Omega \text{Diag}(e^{i\lambda_j}) \Omega^{-1}. \quad (16)$$

In order to finish the calculation, we wish to find a simpler form for the λ_j . we have

$$\lambda_j = \frac{2\pi}{n} \left(\frac{n+1}{2} + \sum_{k=1}^{n-1} \frac{\omega^{k(j+1)}}{\omega^k - 1} \right).$$

We will use an intermediate formula whose proof is below:

Lemma 2. For $n \in \mathbb{N}, j \leq n$, $\sum_{k=1}^{n-1} \frac{\omega^{kj}}{\omega^k - 1} \equiv \frac{n+1}{2} - j \pmod{n}$.

We can thus simplify λ_j :

$$\lambda_j \equiv \frac{2\pi}{n} \left(\frac{n+1}{2} + \frac{n+1}{2} - j - 1 \right) \equiv \frac{2\pi}{n} (-j) \pmod{2\pi},$$

so that $e^{i\lambda_j} = \omega^{-j}$. By inserting it back in Eq.(16), and by using that $(\Omega^{-1})_{jl} = \omega^{-(j-1)(l-1)}/n$, we see that

$$\exp\{iJ\} = \Omega \text{Diag}(\omega^{-j}) \Omega^{-1} = \begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ 1 & 0 & \dots & \dots & 0 \\ 0 & 1 & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix}.$$

In fact (we index from 0 for simpler reading),

$$(\Omega \text{Diag}(\omega^{-j}) \Omega^{-1})_{kl} = \sum_j \omega^{kj} \omega^{-j} \omega^{-jl} / n = \sum_j (\omega^{k-1-l})^j / n = \delta_{k-1,l}.$$

We have thus proven that the operator $\hat{\mathcal{F}}_n^\dagger \hat{\mathcal{R}}_n \hat{\mathcal{F}}_n$ acts on the $\hat{a}_p$ like the permutation matrix, *i.e.* like the cyclic shift operator, so that it equals $\hat{S}_n$.

We will now prove Lemma (2) that we just used *i.e.* that for $n \in \mathbb{N}$, $j \leq n$,

$$\sum_{k=1}^{n-1} \frac{\omega^{kj}}{\omega^k - 1} \equiv \frac{n+1}{2} - j \pmod{n}.$$

Proof. We start by introducing $f(z) = \sum_{k=1}^{n-1} \frac{\omega^{kj}}{1 - z\omega^k}$ for $z \in [0, 1]$, and we are interested in the value of $-f(1)$. For $z < 1$ we can then expand the Taylor series (because $|z\omega^k| < 1$):

$$\begin{aligned} f(z) &= \sum_{k=1}^{n-1} \omega^{kj} \sum_{l=0}^{\infty} z^l \omega^{kl} \\ &= \sum_{l=0}^{\infty} z^l \sum_{k=1}^{n-1} \omega^{(j+l)k}. \end{aligned}$$

We then use the fact that $1 + \sum_{k=1}^{n-1} \omega^{km} = n\mathbb{1}_{n|m}$, so that

$$\begin{aligned} f(z) &= \sum_{l=0}^{\infty} z^l (n\mathbb{1}_{n|j+l} - 1) \\ &= n \left(\delta_{j0} + \sum_{l'=1}^{\infty} z^{l'n-j} \right) - \sum_{l=0}^{\infty} z^l, \end{aligned}$$

where we had to distinguish the case $j = 0$. But because we only care of the result mod n , we can forget the first term δ_{j0} .

$$\begin{aligned} f(z) &\equiv \frac{nz^{n-j}}{1 - z^n} - \frac{1}{1 - z} \pmod{n} \\ &= \frac{nz^{n-j} - (1 + z + z^2 + \dots + z^{n-1})}{1 - z^n}. \end{aligned}$$

We then look at what happens when $z \rightarrow 1$ using L'Hôpital's rule

$$f(1) = \frac{n(n-j) - (0 + 1 + \dots + n-1)}{-n} = -\left(\frac{n+1}{2} - j\right)$$

We then obtain what was claimed: $-f(1) \equiv \frac{n+1}{2} - j$. □

C Parallel project: No-go theorem for Gaussian conversion protocol

In parallel of the Virtual Distillation project, I was given the opportunity to work on a secondary subject, namely "no-go theorem for Gaussian conversion protocols". This project is a lot closer to the main subjects dealt in the group I was in, which focused on continuous variable resource theory.

The goal of this subsidiary project was to prove a no-go theorem formally, as it had been observed numerically on enough inputs to believe so. The main framework is the study of Gaussian operations, *i.e.* a restricted set of operations on a bosonic system, typically easy to perform in current platforms. Starting from the vacuum state, these Gaussian operations can produce any Gaussian state such as displaced or squeezed vacuum, but also multi-mode entangled states. However, if a machine only supports Gaussian states, Gaussian operations and Gaussian measurements (homodyne and heterodyne for example), then it is proven to be efficiently simulatable by a classical computer, thus bringing no exponential quantum advantage.

This leads to the study of non-Gaussianity, *i.e.* of states which do not stem from Gaussian operations on vacuum, which include notably Wigner-negative states (states whose Wigner function takes negative values) such as Fock states $|1\rangle, |2\rangle, \dots$, cat states, etc.

One natural question is what can we do if we are able to produce one such state, is it possible to convert it to another one using only Gaussian operations? While we know that in general this is not possible, there is no clear answer to how close we can get to a target state given an initial state. Moreover, if we are allowed to start with several copies, would this allow for a better precision of the conversion?

Numerical evidence shows that in a simple setup, no benefit could arise from allowing two copies of the initial state as compared to the case of a single copy, and this is what my project was aiming at proving. In Figure 13 are depicted the two setups that we take into account: in the one-mode case we only apply a Gaussian unitary, and in the two-mode we then trace out the first one (by measuring and discarding the outcome), resulting in a mixed state for the second mode if there was some entanglement.



Figure 13: Circuits corresponding to the two kinds of conversions.

The no-go theorem states as follows:

Theorem. For a given initial state $|\psi\rangle$ and target state $|\tau\rangle$, the maximum value of $\langle\tau|\phi\rangle$ over all possible $\hat{U}_G^{(1)}$ is the same as the maximum value of $\langle\tau|\rho|\tau\rangle$ over all possible $\hat{U}_G^{(2)}$.

In order to tackle this problem, I made great use of advanced tools for bosonic systems such as the characteristic function, symplectic transformations and Laguerre polynomials. Despite our effort, we could only prove the theorem for very simple cases and our leads did not find yet a general formulation.